

Assignment 1

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Assignment 1

Problems from R for Data Science book:

Section 1.2.5: Problem 8

```
library(tidyverse)           # loads ggplot2 too

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.2      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(palmerpenguins)

##
## Attaching package: 'palmerpenguins'
##
## The following objects are masked from 'package:datasets':
##
##   penguins, penguins_raw

knitr::opts_chunk$set(echo = TRUE)
library(nycflights13)
library(ggplot2)
library(dplyr)
library(tidyr)
library(knitr)
```

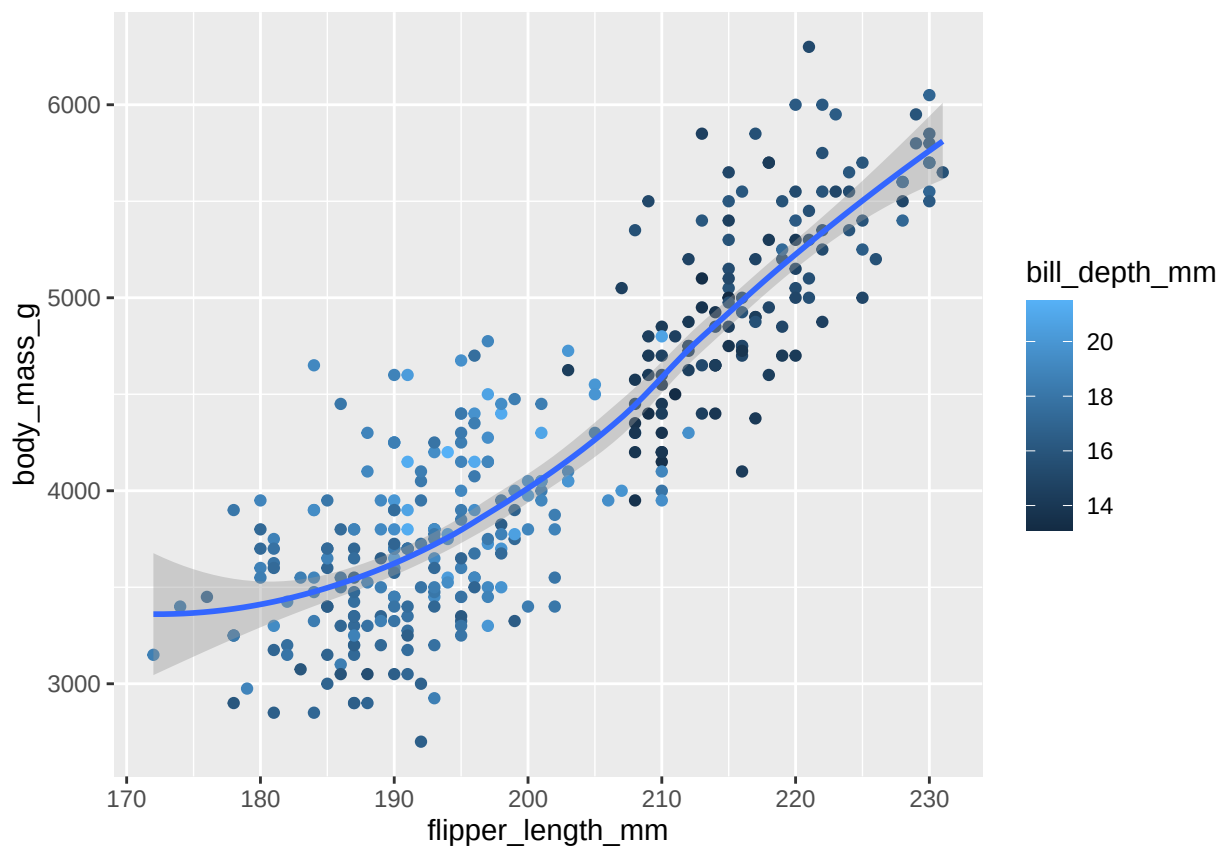
Answer: The code for recreating the visualization is provided below. The `bill_depth_mm` variable should be mapped at the local level, only for the point geom, as it is not used for the smooth geom – the points are colored for bill depth but the smooth line is a single color.

```
ggplot(
  data = penguins,
  aes(x = flipper_length_mm, y = body_mass_g)
) +
  geom_point(aes(color = bill_depth_mm)) +
  geom_smooth()
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Section 1.2.5: Problem 10

No — they'll look the same.

Reason: In the first plot the data and aesthetics are set globally in `ggplot(...)` and inherited by both `geom_point()` and `geom_smooth()`. In the second plot the same data and aesthetics are specified locally for each geom. Since both layers end up with identical data and mappings, the rendered plots are identical. The only difference is code organization; you'd see a difference only if one geom used different data/mappings.

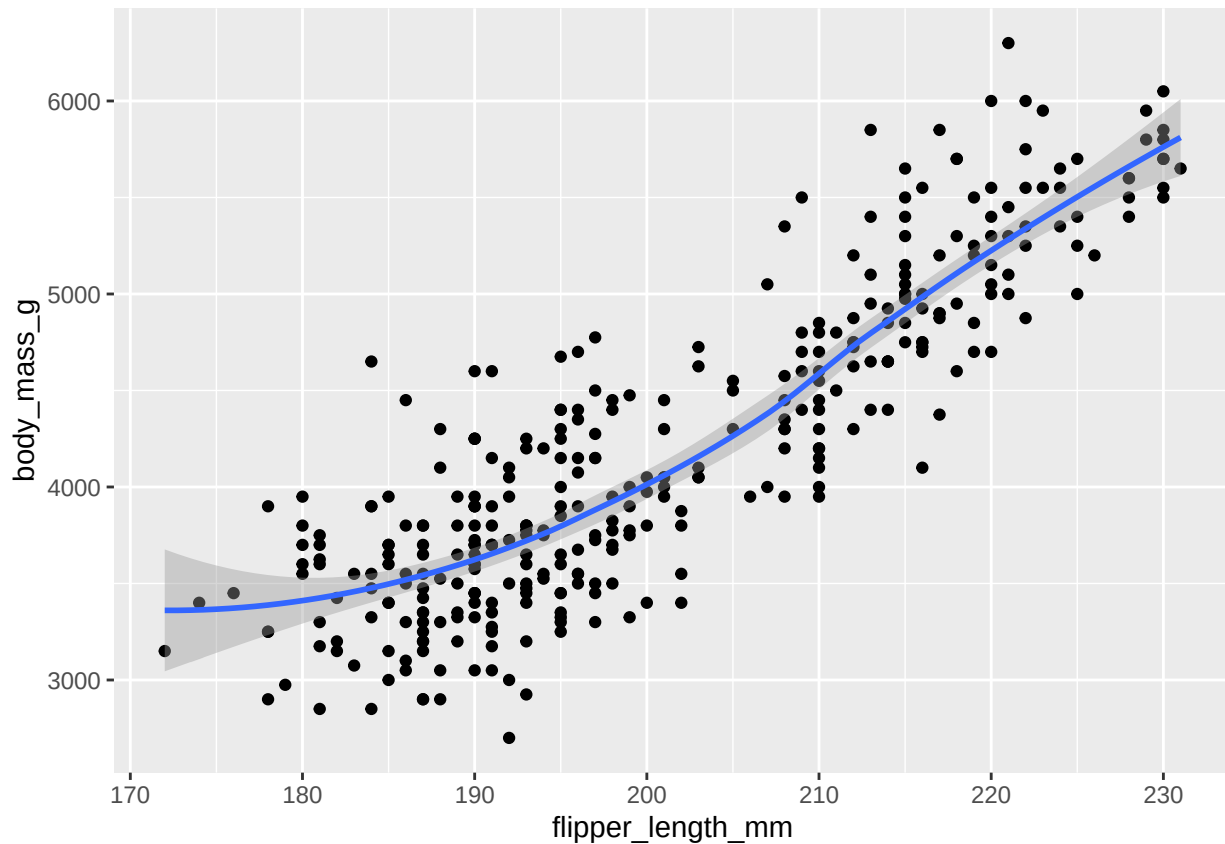
In this plot the data and aesthetics are set globally in `ggplot(...)` and inherited by both `geom_point()` and `geom_smooth()`.

```
ggplot(  
  data = penguins,  
  mapping = aes(x = flipper_length_mm, y = body_mass_g)  
) +  
  geom_point() +  
  geom_smooth()
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range  
## (`stat_smooth()`).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## (`geom_point()`).
```



In this plot the same data and aesthetics are specified locally for each geom.

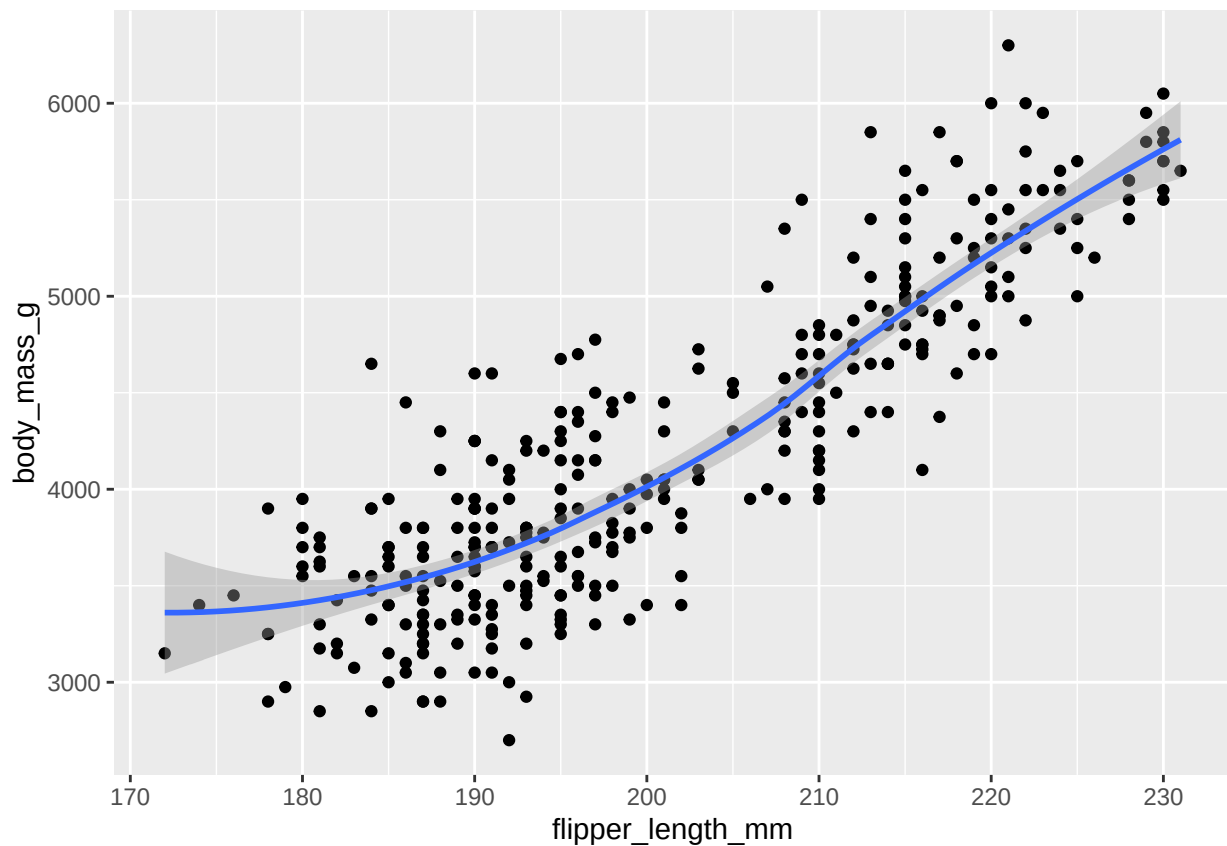
```
ggplot() +  
  geom_point(  
    data = penguins,  
    mapping = aes(x = flipper_length_mm, y = body_mass_g)  
  ) +
```

```
geom_smooth(
  data = penguins,
  mapping = aes(x = flipper_length_mm, y = body_mass_g)
)
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

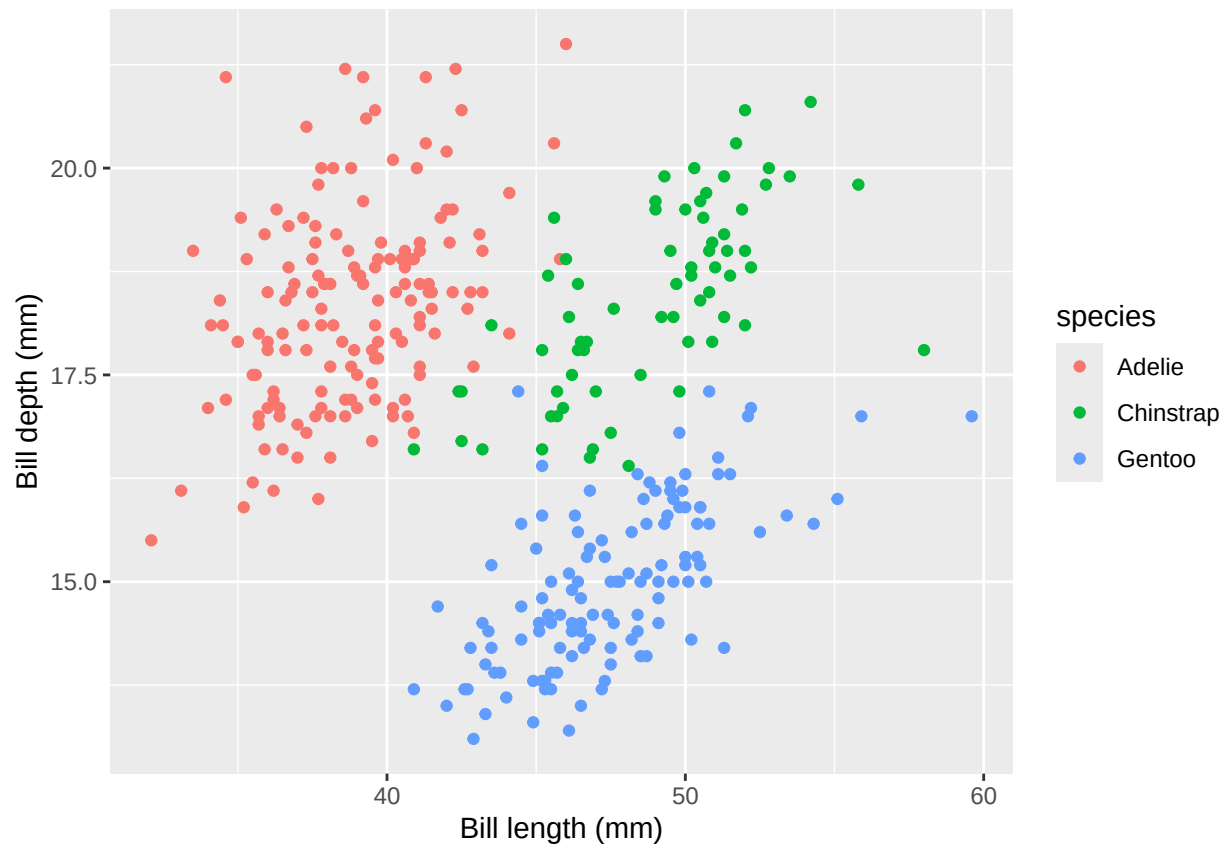


Section 1.5.5: Problem 5

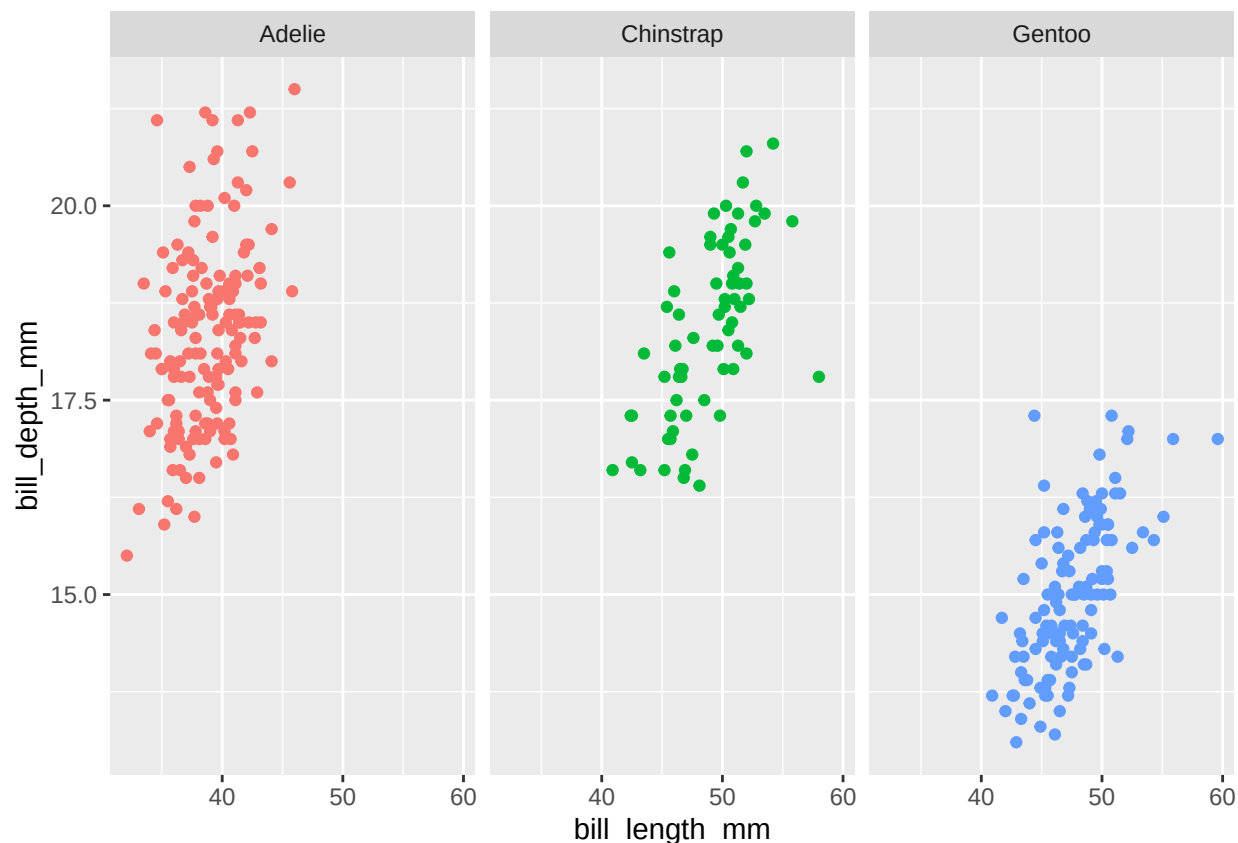
Q. What coloring reveals: Coloring by species shows three clear clusters. The pooled data suggest a negative relationship (longer bills tend to be shallower), but the color reveals this is driven by differences between species (e.g., Gentoo: long & shallow; Adelie: shorter & deeper; Chinstrap: in-between).

Q. What faceting reveals: Faceting separates species into panels, making it clear that within each species the relationship is weak/near-flat (slight positive at most). It highlights that the apparent overall negative trend is largely due to species differences—an example of Simpson's paradox.

```
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm, color = species)) +
  geom_point(na.rm = TRUE) +
  labs(x = "Bill length (mm)", y = "Bill depth (mm)")
```



```
# Faceted by species
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm, color = species)) +
  geom_point(na.rm = TRUE) +
  facet_wrap(~ species) +
  guides(color = "none")
```



Section 3.2.5: Problem 6

Answer: Usually no difference in results, but yes in work done.

Results: If your `filter()` condition doesn't depend on row order, `filter() |> arrange()` and `arrange() |> filter()` yield the same rows in the same order (the filtered rows sorted by your keys). `filter` only removes rows; `arrange` only reorders them.

Work: It's cheaper to filter first. Sorting is $O(n \log n)$; filtering first makes n smaller before the expensive sort.

Exceptions: If your filter depends on order or position (e.g., uses `row_number()`, `lag()/lead()`, `slice_head()`, `distinct(.keep_all=TRUE)` after an `arrange`, etc.), then the order matters for which rows survive.

Section 3.3.5: Problem 5

Answer: Yes—if you typed `contains("TIME")` and still got lowercase `*time` columns, that's expected. By default, select helpers like `contains()`, `starts_with()`, and `ends_with()` are case-insensitive. That's why `contains("TIME")` matches `dep_time`, `arr_time`, `air_time`, `time_hour`, etc. To make matching case-sensitive, set `ignore.case = FALSE`:

```
flights |> select(contains("TIME", ignore.case = FALSE)) # likely returns no columns
```

```
## # A tibble: 336,776 x 0
```

Section 3.5.7: Problem 2

Answer: The following are the top 5 most departure delayed flights from each destination.

```
flights |>
  group_by(dest) |>
  arrange(dest, desc(dep_delay)) |>
  slice_head(n = 5) |>
  relocate(dest, dep_delay)

## # A tibble: 517 x 19
## # Groups:   dest [105]
##   dest dep_delay year month   day dep_time sched_dep_time arr_time
##   <chr>    <dbl> <int> <int> <int>   <int>         <int>    <int>
## 1 ABQ      142  2013    12    14    2223           2001     133
## 2 ABQ      139  2013    12    17    2220           2001     120
## 3 ABQ      125  2013     7    30    2212           2007      57
## 4 ABQ      125  2013     9     2    2212           2007      48
## 5 ABQ      119  2013     7    23    2206           2007     116
## 6 ACK      219  2013     7    23    1139             800    1250
## 7 ACK      138  2013     7     2    1018             800    1119
## 8 ACK      117  2013     7     4     957             800    1106
## 9 ACK      101  2013     5    30    1321           1140    1419
## 10 ACK      100  2013     6    24     940             800    1111
## # i 507 more rows
## # i 11 more variables: sched_arr_time <int>, arr_delay <dbl>, carrier <chr>,
## #   flight <int>, tailnum <chr>, origin <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>
```

Problems from ISL book:

Section 2.4: Problem 2

- a. We collect a set of data on the top 500 firms in the US. For each firm we record profit, number of employees, industry and the CEO salary. We are interested in understanding which factors affect CEO salary.

Ans: $n = 500$, $p = 3$, regression, inference.

- b. We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a success or failure, price charged for the product, marketing budget, competition price, and ten other variables.

Ans: $n = 20$, $p = 13$, classification, prediction.

- c. We are interested in predicting the % change in the USD/Euro exchange rate in relation to the weekly changes in the world stock markets. Hence we collect weekly data for all of 2012. For each week we record the % change in the USD/Euro, the % change in the US market, the % change in the British market, and the % change in the German market.

Ans: $n = 52$, $p = 3$, regression, prediction.

Section 2.4: Problem 4

- (a) Describe three real-life applications in which classification might be useful. Describe the response, as well as the predictors. Is the goal of each application inference or prediction? Explain your answer.

Answer: Classification (Y is categorical) 1. Email spam filtering

Response (Y): spam vs. not spam

Predictors (X): word/character n-grams, sender domain, #links, attachments, time sent

Goal: Prediction — we mainly need to auto-filter incoming mail accurately.

2. Credit-card fraud detection

Response: fraud (yes/no)

Predictors: amount, merchant category, geolocation, device ID, time-of-day, user velocity features

Goal: Prediction — flag suspicious transactions in real time.

3. Patient readmission risk (30-day)

Response: readmitted (yes/no)

Predictors: age, comorbidities, vitals, labs, prior admissions, length of stay, discharge meds

Goal: Prediction — prioritize follow-up and allocate resources; (inference can be secondary to study risk factors).

- (b) Describe three real-life applications in which regression might be useful. Describe the response, as well as the predictors. Is the goal of each application inference or prediction? Explain your answer.

Answer: Regression (Y is numeric/continuous)

House price estimation

Response (Y): sale price (\$)

Predictors (X): square footage, bedrooms, lot size, neighborhood, age, school rating

Goal: Prediction — produce accurate appraisals (inference possible if you want price elasticity wrt size).

Retail demand forecasting

Response: weekly units sold

Predictors: price, promotions, season/holiday flags, competitor price, weather, past sales (lags)

Goal: Prediction — inventory and staffing decisions depend on forecast accuracy.

Flight delay minutes

Response: arrival delay (minutes)

Predictors: carrier, route, scheduled departure time, weather, airport congestion, day of week

Goal: Prediction — estimate delays to inform scheduling and traveler alerts.

- (c) Describe three real-life applications in which cluster analysis might be useful.

Answer: Clustering (unsupervised; no Y)

Customer segmentation

Features: recency/frequency/monetary (RFM), categories browsed, discount sensitivity, session length

Why clustering: group customers into behaviorally similar segments for targeted marketing and personalization.

Document/news grouping

Features: TF-IDF or embeddings of article text, topics, entities, publication patterns

Why clustering: discover topical clusters (e.g., sports, politics, tech) and organize feeds without labels.

Patient phenotype discovery

Features: lab trajectories, vitals, symptoms, imaging-derived features, genomics (if available)

Why clustering: uncover subtypes of a disease to guide tailored treatment and generate new clinical hypotheses.

Section 2.4: Problem 10

(a) To begin, load in the Boston data set. The Boston data set is part of the ISLR2 library.

Answer:

Loading Boston Data set

```
library(ISLR2)
```

data set is contained in the object `Boston`

```
Boston
```

##	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	lstat
## 1	0.00632	18.0	2.31	0	0.5380	6.575	65.2	4.0900	1	296	15.3	4.98
## 2	0.02731	0.0	7.07	0	0.4690	6.421	78.9	4.9671	2	242	17.8	9.14
## 3	0.02729	0.0	7.07	0	0.4690	7.185	61.1	4.9671	2	242	17.8	4.03
## 4	0.03237	0.0	2.18	0	0.4580	6.998	45.8	6.0622	3	222	18.7	2.94
## 5	0.06905	0.0	2.18	0	0.4580	7.147	54.2	6.0622	3	222	18.7	5.33
## 6	0.02985	0.0	2.18	0	0.4580	6.430	58.7	6.0622	3	222	18.7	5.21
## 7	0.08829	12.5	7.87	0	0.5240	6.012	66.6	5.5605	5	311	15.2	12.43
## 8	0.14455	12.5	7.87	0	0.5240	6.172	96.1	5.9505	5	311	15.2	19.15
## 9	0.21124	12.5	7.87	0	0.5240	5.631	100.0	6.0821	5	311	15.2	29.93
## 10	0.17004	12.5	7.87	0	0.5240	6.004	85.9	6.5921	5	311	15.2	17.10
## 11	0.22489	12.5	7.87	0	0.5240	6.377	94.3	6.3467	5	311	15.2	20.45
## 12	0.11747	12.5	7.87	0	0.5240	6.009	82.9	6.2267	5	311	15.2	13.27
## 13	0.09378	12.5	7.87	0	0.5240	5.889	39.0	5.4509	5	311	15.2	15.71
## 14	0.62976	0.0	8.14	0	0.5380	5.949	61.8	4.7075	4	307	21.0	8.26
## 15	0.63796	0.0	8.14	0	0.5380	6.096	84.5	4.4619	4	307	21.0	10.26
## 16	0.62739	0.0	8.14	0	0.5380	5.834	56.5	4.4986	4	307	21.0	8.47
## 17	1.05393	0.0	8.14	0	0.5380	5.935	29.3	4.4986	4	307	21.0	6.58
## 18	0.78420	0.0	8.14	0	0.5380	5.990	81.7	4.2579	4	307	21.0	14.67
## 19	0.80271	0.0	8.14	0	0.5380	5.456	36.6	3.7965	4	307	21.0	11.69
## 20	0.72580	0.0	8.14	0	0.5380	5.727	69.5	3.7965	4	307	21.0	11.28

## 21	1.25179	0.0	8.14	0	0.5380	5.570	98.1	3.7979	4	307	21.0	21.02
## 22	0.85204	0.0	8.14	0	0.5380	5.965	89.2	4.0123	4	307	21.0	13.83
## 23	1.23247	0.0	8.14	0	0.5380	6.142	91.7	3.9769	4	307	21.0	18.72
## 24	0.98843	0.0	8.14	0	0.5380	5.813	100.0	4.0952	4	307	21.0	19.88
## 25	0.75026	0.0	8.14	0	0.5380	5.924	94.1	4.3996	4	307	21.0	16.30
## 26	0.84054	0.0	8.14	0	0.5380	5.599	85.7	4.4546	4	307	21.0	16.51
## 27	0.67191	0.0	8.14	0	0.5380	5.813	90.3	4.6820	4	307	21.0	14.81
## 28	0.95577	0.0	8.14	0	0.5380	6.047	88.8	4.4534	4	307	21.0	17.28
## 29	0.77299	0.0	8.14	0	0.5380	6.495	94.4	4.4547	4	307	21.0	12.80
## 30	1.00245	0.0	8.14	0	0.5380	6.674	87.3	4.2390	4	307	21.0	11.98
## 31	1.13081	0.0	8.14	0	0.5380	5.713	94.1	4.2330	4	307	21.0	22.60
## 32	1.35472	0.0	8.14	0	0.5380	6.072	100.0	4.1750	4	307	21.0	13.04
## 33	1.38799	0.0	8.14	0	0.5380	5.950	82.0	3.9900	4	307	21.0	27.71
## 34	1.15172	0.0	8.14	0	0.5380	5.701	95.0	3.7872	4	307	21.0	18.35
## 35	1.61282	0.0	8.14	0	0.5380	6.096	96.9	3.7598	4	307	21.0	20.34
## 36	0.06417	0.0	5.96	0	0.4990	5.933	68.2	3.3603	5	279	19.2	9.68
## 37	0.09744	0.0	5.96	0	0.4990	5.841	61.4	3.3779	5	279	19.2	11.41
## 38	0.08014	0.0	5.96	0	0.4990	5.850	41.5	3.9342	5	279	19.2	8.77
## 39	0.17505	0.0	5.96	0	0.4990	5.966	30.2	3.8473	5	279	19.2	10.13
## 40	0.02763	75.0	2.95	0	0.4280	6.595	21.8	5.4011	3	252	18.3	4.32
## 41	0.03359	75.0	2.95	0	0.4280	7.024	15.8	5.4011	3	252	18.3	1.98
## 42	0.12744	0.0	6.91	0	0.4480	6.770	2.9	5.7209	3	233	17.9	4.84
## 43	0.14150	0.0	6.91	0	0.4480	6.169	6.6	5.7209	3	233	17.9	5.81
## 44	0.15936	0.0	6.91	0	0.4480	6.211	6.5	5.7209	3	233	17.9	7.44
## 45	0.12269	0.0	6.91	0	0.4480	6.069	40.0	5.7209	3	233	17.9	9.55
## 46	0.17142	0.0	6.91	0	0.4480	5.682	33.8	5.1004	3	233	17.9	10.21
## 47	0.18836	0.0	6.91	0	0.4480	5.786	33.3	5.1004	3	233	17.9	14.15
## 48	0.22927	0.0	6.91	0	0.4480	6.030	85.5	5.6894	3	233	17.9	18.80
## 49	0.25387	0.0	6.91	0	0.4480	5.399	95.3	5.8700	3	233	17.9	30.81
## 50	0.21977	0.0	6.91	0	0.4480	5.602	62.0	6.0877	3	233	17.9	16.20
## 51	0.08873	21.0	5.64	0	0.4390	5.963	45.7	6.8147	4	243	16.8	13.45
## 52	0.04337	21.0	5.64	0	0.4390	6.115	63.0	6.8147	4	243	16.8	9.43
## 53	0.05360	21.0	5.64	0	0.4390	6.511	21.1	6.8147	4	243	16.8	5.28
## 54	0.04981	21.0	5.64	0	0.4390	5.998	21.4	6.8147	4	243	16.8	8.43
## 55	0.01360	75.0	4.00	0	0.4100	5.888	47.6	7.3197	3	469	21.1	14.80
## 56	0.01311	90.0	1.22	0	0.4030	7.249	21.9	8.6966	5	226	17.9	4.81
## 57	0.02055	85.0	0.74	0	0.4100	6.383	35.7	9.1876	2	313	17.3	5.77
## 58	0.01432	100.0	1.32	0	0.4110	6.816	40.5	8.3248	5	256	15.1	3.95
## 59	0.15445	25.0	5.13	0	0.4530	6.145	29.2	7.8148	8	284	19.7	6.86
## 60	0.10328	25.0	5.13	0	0.4530	5.927	47.2	6.9320	8	284	19.7	9.22
## 61	0.14932	25.0	5.13	0	0.4530	5.741	66.2	7.2254	8	284	19.7	13.15
## 62	0.17171	25.0	5.13	0	0.4530	5.966	93.4	6.8185	8	284	19.7	14.44
## 63	0.11027	25.0	5.13	0	0.4530	6.456	67.8	7.2255	8	284	19.7	6.73
## 64	0.12650	25.0	5.13	0	0.4530	6.762	43.4	7.9809	8	284	19.7	9.50
## 65	0.01951	17.5	1.38	0	0.4161	7.104	59.5	9.2229	3	216	18.6	8.05
## 66	0.03584	80.0	3.37	0	0.3980	6.290	17.8	6.6115	4	337	16.1	4.67
## 67	0.04379	80.0	3.37	0	0.3980	5.787	31.1	6.6115	4	337	16.1	10.24
## 68	0.05789	12.5	6.07	0	0.4090	5.878	21.4	6.4980	4	345	18.9	8.10
## 69	0.13554	12.5	6.07	0	0.4090	5.594	36.8	6.4980	4	345	18.9	13.09
## 70	0.12816	12.5	6.07	0	0.4090	5.885	33.0	6.4980	4	345	18.9	8.79
## 71	0.08826	0.0	10.81	0	0.4130	6.417	6.6	5.2873	4	305	19.2	6.72
## 72	0.15876	0.0	10.81	0	0.4130	5.961	17.5	5.2873	4	305	19.2	9.88
## 73	0.09164	0.0	10.81	0	0.4130	6.065	7.8	5.2873	4	305	19.2	5.52
## 74	0.19539	0.0	10.81	0	0.4130	6.245	6.2	5.2873	4	305	19.2	7.54

## 75	0.07896	0.0	12.83	0	0.4370	6.273	6.0	4.2515	5	398	18.7	6.78
## 76	0.09512	0.0	12.83	0	0.4370	6.286	45.0	4.5026	5	398	18.7	8.94
## 77	0.10153	0.0	12.83	0	0.4370	6.279	74.5	4.0522	5	398	18.7	11.97
## 78	0.08707	0.0	12.83	0	0.4370	6.140	45.8	4.0905	5	398	18.7	10.27
## 79	0.05646	0.0	12.83	0	0.4370	6.232	53.7	5.0141	5	398	18.7	12.34
## 80	0.08387	0.0	12.83	0	0.4370	5.874	36.6	4.5026	5	398	18.7	9.10
## 81	0.04113	25.0	4.86	0	0.4260	6.727	33.5	5.4007	4	281	19.0	5.29
## 82	0.04462	25.0	4.86	0	0.4260	6.619	70.4	5.4007	4	281	19.0	7.22
## 83	0.03659	25.0	4.86	0	0.4260	6.302	32.2	5.4007	4	281	19.0	6.72
## 84	0.03551	25.0	4.86	0	0.4260	6.167	46.7	5.4007	4	281	19.0	7.51
## 85	0.05059	0.0	4.49	0	0.4490	6.389	48.0	4.7794	3	247	18.5	9.62
## 86	0.05735	0.0	4.49	0	0.4490	6.630	56.1	4.4377	3	247	18.5	6.53
## 87	0.05188	0.0	4.49	0	0.4490	6.015	45.1	4.4272	3	247	18.5	12.86
## 88	0.07151	0.0	4.49	0	0.4490	6.121	56.8	3.7476	3	247	18.5	8.44
## 89	0.05660	0.0	3.41	0	0.4890	7.007	86.3	3.4217	2	270	17.8	5.50
## 90	0.05302	0.0	3.41	0	0.4890	7.079	63.1	3.4145	2	270	17.8	5.70
## 91	0.04684	0.0	3.41	0	0.4890	6.417	66.1	3.0923	2	270	17.8	8.81
## 92	0.03932	0.0	3.41	0	0.4890	6.405	73.9	3.0921	2	270	17.8	8.20
## 93	0.04203	28.0	15.04	0	0.4640	6.442	53.6	3.6659	4	270	18.2	8.16
## 94	0.02875	28.0	15.04	0	0.4640	6.211	28.9	3.6659	4	270	18.2	6.21
## 95	0.04294	28.0	15.04	0	0.4640	6.249	77.3	3.6150	4	270	18.2	10.59
## 96	0.12204	0.0	2.89	0	0.4450	6.625	57.8	3.4952	2	276	18.0	6.65
## 97	0.11504	0.0	2.89	0	0.4450	6.163	69.6	3.4952	2	276	18.0	11.34
## 98	0.12083	0.0	2.89	0	0.4450	8.069	76.0	3.4952	2	276	18.0	4.21
## 99	0.08187	0.0	2.89	0	0.4450	7.820	36.9	3.4952	2	276	18.0	3.57
## 100	0.06860	0.0	2.89	0	0.4450	7.416	62.5	3.4952	2	276	18.0	6.19
## 101	0.14866	0.0	8.56	0	0.5200	6.727	79.9	2.7778	5	384	20.9	9.42
## 102	0.11432	0.0	8.56	0	0.5200	6.781	71.3	2.8561	5	384	20.9	7.67
## 103	0.22876	0.0	8.56	0	0.5200	6.405	85.4	2.7147	5	384	20.9	10.63
## 104	0.21161	0.0	8.56	0	0.5200	6.137	87.4	2.7147	5	384	20.9	13.44
## 105	0.13960	0.0	8.56	0	0.5200	6.167	90.0	2.4210	5	384	20.9	12.33
## 106	0.13262	0.0	8.56	0	0.5200	5.851	96.7	2.1069	5	384	20.9	16.47
## 107	0.17120	0.0	8.56	0	0.5200	5.836	91.9	2.2110	5	384	20.9	18.66
## 108	0.13117	0.0	8.56	0	0.5200	6.127	85.2	2.1224	5	384	20.9	14.09
## 109	0.12802	0.0	8.56	0	0.5200	6.474	97.1	2.4329	5	384	20.9	12.27
## 110	0.26363	0.0	8.56	0	0.5200	6.229	91.2	2.5451	5	384	20.9	15.55
## 111	0.10793	0.0	8.56	0	0.5200	6.195	54.4	2.7778	5	384	20.9	13.00
## 112	0.10084	0.0	10.01	0	0.5470	6.715	81.6	2.6775	6	432	17.8	10.16
## 113	0.12329	0.0	10.01	0	0.5470	5.913	92.9	2.3534	6	432	17.8	16.21
## 114	0.22212	0.0	10.01	0	0.5470	6.092	95.4	2.5480	6	432	17.8	17.09
## 115	0.14231	0.0	10.01	0	0.5470	6.254	84.2	2.2565	6	432	17.8	10.45
## 116	0.17134	0.0	10.01	0	0.5470	5.928	88.2	2.4631	6	432	17.8	15.76
## 117	0.13158	0.0	10.01	0	0.5470	6.176	72.5	2.7301	6	432	17.8	12.04
## 118	0.15098	0.0	10.01	0	0.5470	6.021	82.6	2.7474	6	432	17.8	10.30
## 119	0.13058	0.0	10.01	0	0.5470	5.872	73.1	2.4775	6	432	17.8	15.37
## 120	0.14476	0.0	10.01	0	0.5470	5.731	65.2	2.7592	6	432	17.8	13.61
## 121	0.06899	0.0	25.65	0	0.5810	5.870	69.7	2.2577	2	188	19.1	14.37
## 122	0.07165	0.0	25.65	0	0.5810	6.004	84.1	2.1974	2	188	19.1	14.27
## 123	0.09299	0.0	25.65	0	0.5810	5.961	92.9	2.0869	2	188	19.1	17.93
## 124	0.15038	0.0	25.65	0	0.5810	5.856	97.0	1.9444	2	188	19.1	25.41
## 125	0.09849	0.0	25.65	0	0.5810	5.879	95.8	2.0063	2	188	19.1	17.58
## 126	0.16902	0.0	25.65	0	0.5810	5.986	88.4	1.9929	2	188	19.1	14.81
## 127	0.38735	0.0	25.65	0	0.5810	5.613	95.6	1.7572	2	188	19.1	27.26
## 128	0.25915	0.0	21.89	0	0.6240	5.693	96.0	1.7883	4	437	21.2	17.19

## 129	0.32543	0.0	21.89	0	0.6240	6.431	98.8	1.8125	4	437	21.2	15.39
## 130	0.88125	0.0	21.89	0	0.6240	5.637	94.7	1.9799	4	437	21.2	18.34
## 131	0.34006	0.0	21.89	0	0.6240	6.458	98.9	2.1185	4	437	21.2	12.60
## 132	1.19294	0.0	21.89	0	0.6240	6.326	97.7	2.2710	4	437	21.2	12.26
## 133	0.59005	0.0	21.89	0	0.6240	6.372	97.9	2.3274	4	437	21.2	11.12
## 134	0.32982	0.0	21.89	0	0.6240	5.822	95.4	2.4699	4	437	21.2	15.03
## 135	0.97617	0.0	21.89	0	0.6240	5.757	98.4	2.3460	4	437	21.2	17.31
## 136	0.55778	0.0	21.89	0	0.6240	6.335	98.2	2.1107	4	437	21.2	16.96
## 137	0.32264	0.0	21.89	0	0.6240	5.942	93.5	1.9669	4	437	21.2	16.90
## 138	0.35233	0.0	21.89	0	0.6240	6.454	98.4	1.8498	4	437	21.2	14.59
## 139	0.24980	0.0	21.89	0	0.6240	5.857	98.2	1.6686	4	437	21.2	21.32
## 140	0.54452	0.0	21.89	0	0.6240	6.151	97.9	1.6687	4	437	21.2	18.46
## 141	0.29090	0.0	21.89	0	0.6240	6.174	93.6	1.6119	4	437	21.2	24.16
## 142	1.62864	0.0	21.89	0	0.6240	5.019	100.0	1.4394	4	437	21.2	34.41
## 143	3.32105	0.0	19.58	1	0.8710	5.403	100.0	1.3216	5	403	14.7	26.82
## 144	4.09740	0.0	19.58	0	0.8710	5.468	100.0	1.4118	5	403	14.7	26.42
## 145	2.77974	0.0	19.58	0	0.8710	4.903	97.8	1.3459	5	403	14.7	29.29
## 146	2.37934	0.0	19.58	0	0.8710	6.130	100.0	1.4191	5	403	14.7	27.80
## 147	2.15505	0.0	19.58	0	0.8710	5.628	100.0	1.5166	5	403	14.7	16.65
## 148	2.36862	0.0	19.58	0	0.8710	4.926	95.7	1.4608	5	403	14.7	29.53
## 149	2.33099	0.0	19.58	0	0.8710	5.186	93.8	1.5296	5	403	14.7	28.32
## 150	2.73397	0.0	19.58	0	0.8710	5.597	94.9	1.5257	5	403	14.7	21.45
## 151	1.65660	0.0	19.58	0	0.8710	6.122	97.3	1.6180	5	403	14.7	14.10
## 152	1.49632	0.0	19.58	0	0.8710	5.404	100.0	1.5916	5	403	14.7	13.28
## 153	1.12658	0.0	19.58	1	0.8710	5.012	88.0	1.6102	5	403	14.7	12.12
## 154	2.14918	0.0	19.58	0	0.8710	5.709	98.5	1.6232	5	403	14.7	15.79
## 155	1.41385	0.0	19.58	1	0.8710	6.129	96.0	1.7494	5	403	14.7	15.12
## 156	3.53501	0.0	19.58	1	0.8710	6.152	82.6	1.7455	5	403	14.7	15.02
## 157	2.44668	0.0	19.58	0	0.8710	5.272	94.0	1.7364	5	403	14.7	16.14
## 158	1.22358	0.0	19.58	0	0.6050	6.943	97.4	1.8773	5	403	14.7	4.59
## 159	1.34284	0.0	19.58	0	0.6050	6.066	100.0	1.7573	5	403	14.7	6.43
## 160	1.42502	0.0	19.58	0	0.8710	6.510	100.0	1.7659	5	403	14.7	7.39
## 161	1.27346	0.0	19.58	1	0.6050	6.250	92.6	1.7984	5	403	14.7	5.50
## 162	1.46336	0.0	19.58	0	0.6050	7.489	90.8	1.9709	5	403	14.7	1.73
## 163	1.83377	0.0	19.58	1	0.6050	7.802	98.2	2.0407	5	403	14.7	1.92
## 164	1.51902	0.0	19.58	1	0.6050	8.375	93.9	2.1620	5	403	14.7	3.32
## 165	2.24236	0.0	19.58	0	0.6050	5.854	91.8	2.4220	5	403	14.7	11.64
## 166	2.92400	0.0	19.58	0	0.6050	6.101	93.0	2.2834	5	403	14.7	9.81
## 167	2.01019	0.0	19.58	0	0.6050	7.929	96.2	2.0459	5	403	14.7	3.70
## 168	1.80028	0.0	19.58	0	0.6050	5.877	79.2	2.4259	5	403	14.7	12.14
## 169	2.30040	0.0	19.58	0	0.6050	6.319	96.1	2.1000	5	403	14.7	11.10
## 170	2.44953	0.0	19.58	0	0.6050	6.402	95.2	2.2625	5	403	14.7	11.32
## 171	1.20742	0.0	19.58	0	0.6050	5.875	94.6	2.4259	5	403	14.7	14.43
## 172	2.31390	0.0	19.58	0	0.6050	5.880	97.3	2.3887	5	403	14.7	12.03
## 173	0.13914	0.0	4.05	0	0.5100	5.572	88.5	2.5961	5	296	16.6	14.69
## 174	0.09178	0.0	4.05	0	0.5100	6.416	84.1	2.6463	5	296	16.6	9.04
## 175	0.08447	0.0	4.05	0	0.5100	5.859	68.7	2.7019	5	296	16.6	9.64
## 176	0.06664	0.0	4.05	0	0.5100	6.546	33.1	3.1323	5	296	16.6	5.33
## 177	0.07022	0.0	4.05	0	0.5100	6.020	47.2	3.5549	5	296	16.6	10.11
## 178	0.05425	0.0	4.05	0	0.5100	6.315	73.4	3.3175	5	296	16.6	6.29
## 179	0.06642	0.0	4.05	0	0.5100	6.860	74.4	2.9153	5	296	16.6	6.92
## 180	0.05780	0.0	2.46	0	0.4880	6.980	58.4	2.8290	3	193	17.8	5.04
## 181	0.06588	0.0	2.46	0	0.4880	7.765	83.3	2.7410	3	193	17.8	7.56
## 182	0.06888	0.0	2.46	0	0.4880	6.144	62.2	2.5979	3	193	17.8	9.45

## 183	0.09103	0.0	2.46	0	0.4880	7.155	92.2	2.7006	3	193	17.8	4.82
## 184	0.10008	0.0	2.46	0	0.4880	6.563	95.6	2.8470	3	193	17.8	5.68
## 185	0.08308	0.0	2.46	0	0.4880	5.604	89.8	2.9879	3	193	17.8	13.98
## 186	0.06047	0.0	2.46	0	0.4880	6.153	68.8	3.2797	3	193	17.8	13.15
## 187	0.05602	0.0	2.46	0	0.4880	7.831	53.6	3.1992	3	193	17.8	4.45
## 188	0.07875	45.0	3.44	0	0.4370	6.782	41.1	3.7886	5	398	15.2	6.68
## 189	0.12579	45.0	3.44	0	0.4370	6.556	29.1	4.5667	5	398	15.2	4.56
## 190	0.08370	45.0	3.44	0	0.4370	7.185	38.9	4.5667	5	398	15.2	5.39
## 191	0.09068	45.0	3.44	0	0.4370	6.951	21.5	6.4798	5	398	15.2	5.10
## 192	0.06911	45.0	3.44	0	0.4370	6.739	30.8	6.4798	5	398	15.2	4.69
## 193	0.08664	45.0	3.44	0	0.4370	7.178	26.3	6.4798	5	398	15.2	2.87
## 194	0.02187	60.0	2.93	0	0.4010	6.800	9.9	6.2196	1	265	15.6	5.03
## 195	0.01439	60.0	2.93	0	0.4010	6.604	18.8	6.2196	1	265	15.6	4.38
## 196	0.01381	80.0	0.46	0	0.4220	7.875	32.0	5.6484	4	255	14.4	2.97
## 197	0.04011	80.0	1.52	0	0.4040	7.287	34.1	7.3090	2	329	12.6	4.08
## 198	0.04666	80.0	1.52	0	0.4040	7.107	36.6	7.3090	2	329	12.6	8.61
## 199	0.03768	80.0	1.52	0	0.4040	7.274	38.3	7.3090	2	329	12.6	6.62
## 200	0.03150	95.0	1.47	0	0.4030	6.975	15.3	7.6534	3	402	17.0	4.56
## 201	0.01778	95.0	1.47	0	0.4030	7.135	13.9	7.6534	3	402	17.0	4.45
## 202	0.03445	82.5	2.03	0	0.4150	6.162	38.4	6.2700	2	348	14.7	7.43
## 203	0.02177	82.5	2.03	0	0.4150	7.610	15.7	6.2700	2	348	14.7	3.11
## 204	0.03510	95.0	2.68	0	0.4161	7.853	33.2	5.1180	4	224	14.7	3.81
## 205	0.02009	95.0	2.68	0	0.4161	8.034	31.9	5.1180	4	224	14.7	2.88
## 206	0.13642	0.0	10.59	0	0.4890	5.891	22.3	3.9454	4	277	18.6	10.87
## 207	0.22969	0.0	10.59	0	0.4890	6.326	52.5	4.3549	4	277	18.6	10.97
## 208	0.25199	0.0	10.59	0	0.4890	5.783	72.7	4.3549	4	277	18.6	18.06
## 209	0.13587	0.0	10.59	1	0.4890	6.064	59.1	4.2392	4	277	18.6	14.66
## 210	0.43571	0.0	10.59	1	0.4890	5.344	100.0	3.8750	4	277	18.6	23.09
## 211	0.17446	0.0	10.59	1	0.4890	5.960	92.1	3.8771	4	277	18.6	17.27
## 212	0.37578	0.0	10.59	1	0.4890	5.404	88.6	3.6650	4	277	18.6	23.98
## 213	0.21719	0.0	10.59	1	0.4890	5.807	53.8	3.6526	4	277	18.6	16.03
## 214	0.14052	0.0	10.59	0	0.4890	6.375	32.3	3.9454	4	277	18.6	9.38
## 215	0.28955	0.0	10.59	0	0.4890	5.412	9.8	3.5875	4	277	18.6	29.55
## 216	0.19802	0.0	10.59	0	0.4890	6.182	42.4	3.9454	4	277	18.6	9.47
## 217	0.04560	0.0	13.89	1	0.5500	5.888	56.0	3.1121	5	276	16.4	13.51
## 218	0.07013	0.0	13.89	0	0.5500	6.642	85.1	3.4211	5	276	16.4	9.69
## 219	0.11069	0.0	13.89	1	0.5500	5.951	93.8	2.8893	5	276	16.4	17.92
## 220	0.11425	0.0	13.89	1	0.5500	6.373	92.4	3.3633	5	276	16.4	10.50
## 221	0.35809	0.0	6.20	1	0.5070	6.951	88.5	2.8617	8	307	17.4	9.71
## 222	0.40771	0.0	6.20	1	0.5070	6.164	91.3	3.0480	8	307	17.4	21.46
## 223	0.62356	0.0	6.20	1	0.5070	6.879	77.7	3.2721	8	307	17.4	9.93
## 224	0.61470	0.0	6.20	0	0.5070	6.618	80.8	3.2721	8	307	17.4	7.60
## 225	0.31533	0.0	6.20	0	0.5040	8.266	78.3	2.8944	8	307	17.4	4.14
## 226	0.52693	0.0	6.20	0	0.5040	8.725	83.0	2.8944	8	307	17.4	4.63
## 227	0.38214	0.0	6.20	0	0.5040	8.040	86.5	3.2157	8	307	17.4	3.13
## 228	0.41238	0.0	6.20	0	0.5040	7.163	79.9	3.2157	8	307	17.4	6.36
## 229	0.29819	0.0	6.20	0	0.5040	7.686	17.0	3.3751	8	307	17.4	3.92
## 230	0.44178	0.0	6.20	0	0.5040	6.552	21.4	3.3751	8	307	17.4	3.76
## 231	0.53700	0.0	6.20	0	0.5040	5.981	68.1	3.6715	8	307	17.4	11.65
## 232	0.46296	0.0	6.20	0	0.5040	7.412	76.9	3.6715	8	307	17.4	5.25
## 233	0.57529	0.0	6.20	0	0.5070	8.337	73.3	3.8384	8	307	17.4	2.47
## 234	0.33147	0.0	6.20	0	0.5070	8.247	70.4	3.6519	8	307	17.4	3.95
## 235	0.44791	0.0	6.20	1	0.5070	6.726	66.5	3.6519	8	307	17.4	8.05
## 236	0.33045	0.0	6.20	0	0.5070	6.086	61.5	3.6519	8	307	17.4	10.88

## 237	0.52058	0.0	6.20	1	0.5070	6.631	76.5	4.1480	8	307	17.4	9.54
## 238	0.51183	0.0	6.20	0	0.5070	7.358	71.6	4.1480	8	307	17.4	4.73
## 239	0.08244	30.0	4.93	0	0.4280	6.481	18.5	6.1899	6	300	16.6	6.36
## 240	0.09252	30.0	4.93	0	0.4280	6.606	42.2	6.1899	6	300	16.6	7.37
## 241	0.11329	30.0	4.93	0	0.4280	6.897	54.3	6.3361	6	300	16.6	11.38
## 242	0.10612	30.0	4.93	0	0.4280	6.095	65.1	6.3361	6	300	16.6	12.40
## 243	0.10290	30.0	4.93	0	0.4280	6.358	52.9	7.0355	6	300	16.6	11.22
## 244	0.12757	30.0	4.93	0	0.4280	6.393	7.8	7.0355	6	300	16.6	5.19
## 245	0.20608	22.0	5.86	0	0.4310	5.593	76.5	7.9549	7	330	19.1	12.50
## 246	0.19133	22.0	5.86	0	0.4310	5.605	70.2	7.9549	7	330	19.1	18.46
## 247	0.33983	22.0	5.86	0	0.4310	6.108	34.9	8.0555	7	330	19.1	9.16
## 248	0.19657	22.0	5.86	0	0.4310	6.226	79.2	8.0555	7	330	19.1	10.15
## 249	0.16439	22.0	5.86	0	0.4310	6.433	49.1	7.8265	7	330	19.1	9.52
## 250	0.19073	22.0	5.86	0	0.4310	6.718	17.5	7.8265	7	330	19.1	6.56
## 251	0.14030	22.0	5.86	0	0.4310	6.487	13.0	7.3967	7	330	19.1	5.90
## 252	0.21409	22.0	5.86	0	0.4310	6.438	8.9	7.3967	7	330	19.1	3.59
## 253	0.08221	22.0	5.86	0	0.4310	6.957	6.8	8.9067	7	330	19.1	3.53
## 254	0.36894	22.0	5.86	0	0.4310	8.259	8.4	8.9067	7	330	19.1	3.54
## 255	0.04819	80.0	3.64	0	0.3920	6.108	32.0	9.2203	1	315	16.4	6.57
## 256	0.03548	80.0	3.64	0	0.3920	5.876	19.1	9.2203	1	315	16.4	9.25
## 257	0.01538	90.0	3.75	0	0.3940	7.454	34.2	6.3361	3	244	15.9	3.11
## 258	0.61154	20.0	3.97	0	0.6470	8.704	86.9	1.8010	5	264	13.0	5.12
## 259	0.66351	20.0	3.97	0	0.6470	7.333	100.0	1.8946	5	264	13.0	7.79
## 260	0.65665	20.0	3.97	0	0.6470	6.842	100.0	2.0107	5	264	13.0	6.90
## 261	0.54011	20.0	3.97	0	0.6470	7.203	81.8	2.1121	5	264	13.0	9.59
## 262	0.53412	20.0	3.97	0	0.6470	7.520	89.4	2.1398	5	264	13.0	7.26
## 263	0.52014	20.0	3.97	0	0.6470	8.398	91.5	2.2885	5	264	13.0	5.91
## 264	0.82526	20.0	3.97	0	0.6470	7.327	94.5	2.0788	5	264	13.0	11.25
## 265	0.55007	20.0	3.97	0	0.6470	7.206	91.6	1.9301	5	264	13.0	8.10
## 266	0.76162	20.0	3.97	0	0.6470	5.560	62.8	1.9865	5	264	13.0	10.45
## 267	0.78570	20.0	3.97	0	0.6470	7.014	84.6	2.1329	5	264	13.0	14.79
## 268	0.57834	20.0	3.97	0	0.5750	8.297	67.0	2.4216	5	264	13.0	7.44
## 269	0.54050	20.0	3.97	0	0.5750	7.470	52.6	2.8720	5	264	13.0	3.16
## 270	0.09065	20.0	6.96	1	0.4640	5.920	61.5	3.9175	3	223	18.6	13.65
## 271	0.29916	20.0	6.96	0	0.4640	5.856	42.1	4.4290	3	223	18.6	13.00
## 272	0.16211	20.0	6.96	0	0.4640	6.240	16.3	4.4290	3	223	18.6	6.59
## 273	0.11460	20.0	6.96	0	0.4640	6.538	58.7	3.9175	3	223	18.6	7.73
## 274	0.22188	20.0	6.96	1	0.4640	7.691	51.8	4.3665	3	223	18.6	6.58
## 275	0.05644	40.0	6.41	1	0.4470	6.758	32.9	4.0776	4	254	17.6	3.53
## 276	0.09604	40.0	6.41	0	0.4470	6.854	42.8	4.2673	4	254	17.6	2.98
## 277	0.10469	40.0	6.41	1	0.4470	7.267	49.0	4.7872	4	254	17.6	6.05
## 278	0.06127	40.0	6.41	1	0.4470	6.826	27.6	4.8628	4	254	17.6	4.16
## 279	0.07978	40.0	6.41	0	0.4470	6.482	32.1	4.1403	4	254	17.6	7.19
## 280	0.21038	20.0	3.33	0	0.4429	6.812	32.2	4.1007	5	216	14.9	4.85
## 281	0.03578	20.0	3.33	0	0.4429	7.820	64.5	4.6947	5	216	14.9	3.76
## 282	0.03705	20.0	3.33	0	0.4429	6.968	37.2	5.2447	5	216	14.9	4.59
## 283	0.06129	20.0	3.33	1	0.4429	7.645	49.7	5.2119	5	216	14.9	3.01
## 284	0.01501	90.0	1.21	1	0.4010	7.923	24.8	5.8850	1	198	13.6	3.16
## 285	0.00906	90.0	2.97	0	0.4000	7.088	20.8	7.3073	1	285	15.3	7.85
## 286	0.01096	55.0	2.25	0	0.3890	6.453	31.9	7.3073	1	300	15.3	8.23
## 287	0.01965	80.0	1.76	0	0.3850	6.230	31.5	9.0892	1	241	18.2	12.93
## 288	0.03871	52.5	5.32	0	0.4050	6.209	31.3	7.3172	6	293	16.6	7.14
## 289	0.04590	52.5	5.32	0	0.4050	6.315	45.6	7.3172	6	293	16.6	7.60
## 290	0.04297	52.5	5.32	0	0.4050	6.565	22.9	7.3172	6	293	16.6	9.51

## 291	0.03502	80.0	4.95	0 0.4110	6.861	27.9	5.1167	4 245	19.2	3.33
## 292	0.07886	80.0	4.95	0 0.4110	7.148	27.7	5.1167	4 245	19.2	3.56
## 293	0.03615	80.0	4.95	0 0.4110	6.630	23.4	5.1167	4 245	19.2	4.70
## 294	0.08265	0.0	13.92	0 0.4370	6.127	18.4	5.5027	4 289	16.0	8.58
## 295	0.08199	0.0	13.92	0 0.4370	6.009	42.3	5.5027	4 289	16.0	10.40
## 296	0.12932	0.0	13.92	0 0.4370	6.678	31.1	5.9604	4 289	16.0	6.27
## 297	0.05372	0.0	13.92	0 0.4370	6.549	51.0	5.9604	4 289	16.0	7.39
## 298	0.14103	0.0	13.92	0 0.4370	5.790	58.0	6.3200	4 289	16.0	15.84
## 299	0.06466	70.0	2.24	0 0.4000	6.345	20.1	7.8278	5 358	14.8	4.97
## 300	0.05561	70.0	2.24	0 0.4000	7.041	10.0	7.8278	5 358	14.8	4.74
## 301	0.04417	70.0	2.24	0 0.4000	6.871	47.4	7.8278	5 358	14.8	6.07
## 302	0.03537	34.0	6.09	0 0.4330	6.590	40.4	5.4917	7 329	16.1	9.50
## 303	0.09266	34.0	6.09	0 0.4330	6.495	18.4	5.4917	7 329	16.1	8.67
## 304	0.10000	34.0	6.09	0 0.4330	6.982	17.7	5.4917	7 329	16.1	4.86
## 305	0.05515	33.0	2.18	0 0.4720	7.236	41.1	4.0220	7 222	18.4	6.93
## 306	0.05479	33.0	2.18	0 0.4720	6.616	58.1	3.3700	7 222	18.4	8.93
## 307	0.07503	33.0	2.18	0 0.4720	7.420	71.9	3.0992	7 222	18.4	6.47
## 308	0.04932	33.0	2.18	0 0.4720	6.849	70.3	3.1827	7 222	18.4	7.53
## 309	0.49298	0.0	9.90	0 0.5440	6.635	82.5	3.3175	4 304	18.4	4.54
## 310	0.34940	0.0	9.90	0 0.5440	5.972	76.7	3.1025	4 304	18.4	9.97
## 311	2.63548	0.0	9.90	0 0.5440	4.973	37.8	2.5194	4 304	18.4	12.64
## 312	0.79041	0.0	9.90	0 0.5440	6.122	52.8	2.6403	4 304	18.4	5.98
## 313	0.26169	0.0	9.90	0 0.5440	6.023	90.4	2.8340	4 304	18.4	11.72
## 314	0.26938	0.0	9.90	0 0.5440	6.266	82.8	3.2628	4 304	18.4	7.90
## 315	0.36920	0.0	9.90	0 0.5440	6.567	87.3	3.6023	4 304	18.4	9.28
## 316	0.25356	0.0	9.90	0 0.5440	5.705	77.7	3.9450	4 304	18.4	11.50
## 317	0.31827	0.0	9.90	0 0.5440	5.914	83.2	3.9986	4 304	18.4	18.33
## 318	0.24522	0.0	9.90	0 0.5440	5.782	71.7	4.0317	4 304	18.4	15.94
## 319	0.40202	0.0	9.90	0 0.5440	6.382	67.2	3.5325	4 304	18.4	10.36
## 320	0.47547	0.0	9.90	0 0.5440	6.113	58.8	4.0019	4 304	18.4	12.73
## 321	0.16760	0.0	7.38	0 0.4930	6.426	52.3	4.5404	5 287	19.6	7.20
## 322	0.18159	0.0	7.38	0 0.4930	6.376	54.3	4.5404	5 287	19.6	6.87
## 323	0.35114	0.0	7.38	0 0.4930	6.041	49.9	4.7211	5 287	19.6	7.70
## 324	0.28392	0.0	7.38	0 0.4930	5.708	74.3	4.7211	5 287	19.6	11.74
## 325	0.34109	0.0	7.38	0 0.4930	6.415	40.1	4.7211	5 287	19.6	6.12
## 326	0.19186	0.0	7.38	0 0.4930	6.431	14.7	5.4159	5 287	19.6	5.08
## 327	0.30347	0.0	7.38	0 0.4930	6.312	28.9	5.4159	5 287	19.6	6.15
## 328	0.24103	0.0	7.38	0 0.4930	6.083	43.7	5.4159	5 287	19.6	12.79
## 329	0.06617	0.0	3.24	0 0.4600	5.868	25.8	5.2146	4 430	16.9	9.97
## 330	0.06724	0.0	3.24	0 0.4600	6.333	17.2	5.2146	4 430	16.9	7.34
## 331	0.04544	0.0	3.24	0 0.4600	6.144	32.2	5.8736	4 430	16.9	9.09
## 332	0.05023	35.0	6.06	0 0.4379	5.706	28.4	6.6407	1 304	16.9	12.43
## 333	0.03466	35.0	6.06	0 0.4379	6.031	23.3	6.6407	1 304	16.9	7.83
## 334	0.05083	0.0	5.19	0 0.5150	6.316	38.1	6.4584	5 224	20.2	5.68
## 335	0.03738	0.0	5.19	0 0.5150	6.310	38.5	6.4584	5 224	20.2	6.75
## 336	0.03961	0.0	5.19	0 0.5150	6.037	34.5	5.9853	5 224	20.2	8.01
## 337	0.03427	0.0	5.19	0 0.5150	5.869	46.3	5.2311	5 224	20.2	9.80
## 338	0.03041	0.0	5.19	0 0.5150	5.895	59.6	5.6150	5 224	20.2	10.56
## 339	0.03306	0.0	5.19	0 0.5150	6.059	37.3	4.8122	5 224	20.2	8.51
## 340	0.05497	0.0	5.19	0 0.5150	5.985	45.4	4.8122	5 224	20.2	9.74
## 341	0.06151	0.0	5.19	0 0.5150	5.968	58.5	4.8122	5 224	20.2	9.29
## 342	0.01301	35.0	1.52	0 0.4420	7.241	49.3	7.0379	1 284	15.5	5.49
## 343	0.02498	0.0	1.89	0 0.5180	6.540	59.7	6.2669	1 422	15.9	8.65
## 344	0.02543	55.0	3.78	0 0.4840	6.696	56.4	5.7321	5 370	17.6	7.18

## 345	0.03049	55.0	3.78	0 0.4840	6.874	28.1	6.4654	5 370	17.6	4.61
## 346	0.03113	0.0	4.39	0 0.4420	6.014	48.5	8.0136	3 352	18.8	10.53
## 347	0.06162	0.0	4.39	0 0.4420	5.898	52.3	8.0136	3 352	18.8	12.67
## 348	0.01870	85.0	4.15	0 0.4290	6.516	27.7	8.5353	4 351	17.9	6.36
## 349	0.01501	80.0	2.01	0 0.4350	6.635	29.7	8.3440	4 280	17.0	5.99
## 350	0.02899	40.0	1.25	0 0.4290	6.939	34.5	8.7921	1 335	19.7	5.89
## 351	0.06211	40.0	1.25	0 0.4290	6.490	44.4	8.7921	1 335	19.7	5.98
## 352	0.07950	60.0	1.69	0 0.4110	6.579	35.9	10.7103	4 411	18.3	5.49
## 353	0.07244	60.0	1.69	0 0.4110	5.884	18.5	10.7103	4 411	18.3	7.79
## 354	0.01709	90.0	2.02	0 0.4100	6.728	36.1	12.1265	5 187	17.0	4.50
## 355	0.04301	80.0	1.91	0 0.4130	5.663	21.9	10.5857	4 334	22.0	8.05
## 356	0.10659	80.0	1.91	0 0.4130	5.936	19.5	10.5857	4 334	22.0	5.57
## 357	8.98296	0.0	18.10	1 0.7700	6.212	97.4	2.1222	24 666	20.2	17.60
## 358	3.84970	0.0	18.10	1 0.7700	6.395	91.0	2.5052	24 666	20.2	13.27
## 359	5.20177	0.0	18.10	1 0.7700	6.127	83.4	2.7227	24 666	20.2	11.48
## 360	4.26131	0.0	18.10	0 0.7700	6.112	81.3	2.5091	24 666	20.2	12.67
## 361	4.54192	0.0	18.10	0 0.7700	6.398	88.0	2.5182	24 666	20.2	7.79
## 362	3.83684	0.0	18.10	0 0.7700	6.251	91.1	2.2955	24 666	20.2	14.19
## 363	3.67822	0.0	18.10	0 0.7700	5.362	96.2	2.1036	24 666	20.2	10.19
## 364	4.22239	0.0	18.10	1 0.7700	5.803	89.0	1.9047	24 666	20.2	14.64
## 365	3.47428	0.0	18.10	1 0.7180	8.780	82.9	1.9047	24 666	20.2	5.29
## 366	4.55587	0.0	18.10	0 0.7180	3.561	87.9	1.6132	24 666	20.2	7.12
## 367	3.69695	0.0	18.10	0 0.7180	4.963	91.4	1.7523	24 666	20.2	14.00
## 368	13.52220	0.0	18.10	0 0.6310	3.863	100.0	1.5106	24 666	20.2	13.33
## 369	4.89822	0.0	18.10	0 0.6310	4.970	100.0	1.3325	24 666	20.2	3.26
## 370	5.66998	0.0	18.10	1 0.6310	6.683	96.8	1.3567	24 666	20.2	3.73
## 371	6.53876	0.0	18.10	1 0.6310	7.016	97.5	1.2024	24 666	20.2	2.96
## 372	9.23230	0.0	18.10	0 0.6310	6.216	100.0	1.1691	24 666	20.2	9.53
## 373	8.26725	0.0	18.10	1 0.6680	5.875	89.6	1.1296	24 666	20.2	8.88
## 374	11.10810	0.0	18.10	0 0.6680	4.906	100.0	1.1742	24 666	20.2	34.77
## 375	18.49820	0.0	18.10	0 0.6680	4.138	100.0	1.1370	24 666	20.2	37.97
## 376	19.60910	0.0	18.10	0 0.6710	7.313	97.9	1.3163	24 666	20.2	13.44
## 377	15.28800	0.0	18.10	0 0.6710	6.649	93.3	1.3449	24 666	20.2	23.24
## 378	9.82349	0.0	18.10	0 0.6710	6.794	98.8	1.3580	24 666	20.2	21.24
## 379	23.64820	0.0	18.10	0 0.6710	6.380	96.2	1.3861	24 666	20.2	23.69
## 380	17.86670	0.0	18.10	0 0.6710	6.223	100.0	1.3861	24 666	20.2	21.78
## 381	88.97620	0.0	18.10	0 0.6710	6.968	91.9	1.4165	24 666	20.2	17.21
## 382	15.87440	0.0	18.10	0 0.6710	6.545	99.1	1.5192	24 666	20.2	21.08
## 383	9.18702	0.0	18.10	0 0.7000	5.536	100.0	1.5804	24 666	20.2	23.60
## 384	7.99248	0.0	18.10	0 0.7000	5.520	100.0	1.5331	24 666	20.2	24.56
## 385	20.08490	0.0	18.10	0 0.7000	4.368	91.2	1.4395	24 666	20.2	30.63
## 386	16.81180	0.0	18.10	0 0.7000	5.277	98.1	1.4261	24 666	20.2	30.81
## 387	24.39380	0.0	18.10	0 0.7000	4.652	100.0	1.4672	24 666	20.2	28.28
## 388	22.59710	0.0	18.10	0 0.7000	5.000	89.5	1.5184	24 666	20.2	31.99
## 389	14.33370	0.0	18.10	0 0.7000	4.880	100.0	1.5895	24 666	20.2	30.62
## 390	8.15174	0.0	18.10	0 0.7000	5.390	98.9	1.7281	24 666	20.2	20.85
## 391	6.96215	0.0	18.10	0 0.7000	5.713	97.0	1.9265	24 666	20.2	17.11
## 392	5.29305	0.0	18.10	0 0.7000	6.051	82.5	2.1678	24 666	20.2	18.76
## 393	11.57790	0.0	18.10	0 0.7000	5.036	97.0	1.7700	24 666	20.2	25.68
## 394	8.64476	0.0	18.10	0 0.6930	6.193	92.6	1.7912	24 666	20.2	15.17
## 395	13.35980	0.0	18.10	0 0.6930	5.887	94.7	1.7821	24 666	20.2	16.35
## 396	8.71675	0.0	18.10	0 0.6930	6.471	98.8	1.7257	24 666	20.2	17.12
## 397	5.87205	0.0	18.10	0 0.6930	6.405	96.0	1.6768	24 666	20.2	19.37
## 398	7.67202	0.0	18.10	0 0.6930	5.747	98.9	1.6334	24 666	20.2	19.92

## 399	38.35180	0.0	18.10	0	0.6930	5.453	100.0	1.4896	24	666	20.2	30.59
## 400	9.91655	0.0	18.10	0	0.6930	5.852	77.8	1.5004	24	666	20.2	29.97
## 401	25.04610	0.0	18.10	0	0.6930	5.987	100.0	1.5888	24	666	20.2	26.77
## 402	14.23620	0.0	18.10	0	0.6930	6.343	100.0	1.5741	24	666	20.2	20.32
## 403	9.59571	0.0	18.10	0	0.6930	6.404	100.0	1.6390	24	666	20.2	20.31
## 404	24.80170	0.0	18.10	0	0.6930	5.349	96.0	1.7028	24	666	20.2	19.77
## 405	41.52920	0.0	18.10	0	0.6930	5.531	85.4	1.6074	24	666	20.2	27.38
## 406	67.92080	0.0	18.10	0	0.6930	5.683	100.0	1.4254	24	666	20.2	22.98
## 407	20.71620	0.0	18.10	0	0.6590	4.138	100.0	1.1781	24	666	20.2	23.34
## 408	11.95110	0.0	18.10	0	0.6590	5.608	100.0	1.2852	24	666	20.2	12.13
## 409	7.40389	0.0	18.10	0	0.5970	5.617	97.9	1.4547	24	666	20.2	26.40
## 410	14.43830	0.0	18.10	0	0.5970	6.852	100.0	1.4655	24	666	20.2	19.78
## 411	51.13580	0.0	18.10	0	0.5970	5.757	100.0	1.4130	24	666	20.2	10.11
## 412	14.05070	0.0	18.10	0	0.5970	6.657	100.0	1.5275	24	666	20.2	21.22
## 413	18.81100	0.0	18.10	0	0.5970	4.628	100.0	1.5539	24	666	20.2	34.37
## 414	28.65580	0.0	18.10	0	0.5970	5.155	100.0	1.5894	24	666	20.2	20.08
## 415	45.74610	0.0	18.10	0	0.6930	4.519	100.0	1.6582	24	666	20.2	36.98
## 416	18.08460	0.0	18.10	0	0.6790	6.434	100.0	1.8347	24	666	20.2	29.05
## 417	10.83420	0.0	18.10	0	0.6790	6.782	90.8	1.8195	24	666	20.2	25.79
## 418	25.94060	0.0	18.10	0	0.6790	5.304	89.1	1.6475	24	666	20.2	26.64
## 419	73.53410	0.0	18.10	0	0.6790	5.957	100.0	1.8026	24	666	20.2	20.62
## 420	11.81230	0.0	18.10	0	0.7180	6.824	76.5	1.7940	24	666	20.2	22.74
## 421	11.08740	0.0	18.10	0	0.7180	6.411	100.0	1.8589	24	666	20.2	15.02
## 422	7.02259	0.0	18.10	0	0.7180	6.006	95.3	1.8746	24	666	20.2	15.70
## 423	12.04820	0.0	18.10	0	0.6140	5.648	87.6	1.9512	24	666	20.2	14.10
## 424	7.05042	0.0	18.10	0	0.6140	6.103	85.1	2.0218	24	666	20.2	23.29
## 425	8.79212	0.0	18.10	0	0.5840	5.565	70.6	2.0635	24	666	20.2	17.16
## 426	15.86030	0.0	18.10	0	0.6790	5.896	95.4	1.9096	24	666	20.2	24.39
## 427	12.24720	0.0	18.10	0	0.5840	5.837	59.7	1.9976	24	666	20.2	15.69
## 428	37.66190	0.0	18.10	0	0.6790	6.202	78.7	1.8629	24	666	20.2	14.52
## 429	7.36711	0.0	18.10	0	0.6790	6.193	78.1	1.9356	24	666	20.2	21.52
## 430	9.33889	0.0	18.10	0	0.6790	6.380	95.6	1.9682	24	666	20.2	24.08
## 431	8.49213	0.0	18.10	0	0.5840	6.348	86.1	2.0527	24	666	20.2	17.64
## 432	10.06230	0.0	18.10	0	0.5840	6.833	94.3	2.0882	24	666	20.2	19.69
## 433	6.44405	0.0	18.10	0	0.5840	6.425	74.8	2.2004	24	666	20.2	12.03
## 434	5.58107	0.0	18.10	0	0.7130	6.436	87.9	2.3158	24	666	20.2	16.22
## 435	13.91340	0.0	18.10	0	0.7130	6.208	95.0	2.2222	24	666	20.2	15.17
## 436	11.16040	0.0	18.10	0	0.7400	6.629	94.6	2.1247	24	666	20.2	23.27
## 437	14.42080	0.0	18.10	0	0.7400	6.461	93.3	2.0026	24	666	20.2	18.05
## 438	15.17720	0.0	18.10	0	0.7400	6.152	100.0	1.9142	24	666	20.2	26.45
## 439	13.67810	0.0	18.10	0	0.7400	5.935	87.9	1.8206	24	666	20.2	34.02
## 440	9.39063	0.0	18.10	0	0.7400	5.627	93.9	1.8172	24	666	20.2	22.88
## 441	22.05110	0.0	18.10	0	0.7400	5.818	92.4	1.8662	24	666	20.2	22.11
## 442	9.72418	0.0	18.10	0	0.7400	6.406	97.2	2.0651	24	666	20.2	19.52
## 443	5.66637	0.0	18.10	0	0.7400	6.219	100.0	2.0048	24	666	20.2	16.59
## 444	9.96654	0.0	18.10	0	0.7400	6.485	100.0	1.9784	24	666	20.2	18.85
## 445	12.80230	0.0	18.10	0	0.7400	5.854	96.6	1.8956	24	666	20.2	23.79
## 446	10.67180	0.0	18.10	0	0.7400	6.459	94.8	1.9879	24	666	20.2	23.98
## 447	6.28807	0.0	18.10	0	0.7400	6.341	96.4	2.0720	24	666	20.2	17.79
## 448	9.92485	0.0	18.10	0	0.7400	6.251	96.6	2.1980	24	666	20.2	16.44
## 449	9.32909	0.0	18.10	0	0.7130	6.185	98.7	2.2616	24	666	20.2	18.13
## 450	7.52601	0.0	18.10	0	0.7130	6.417	98.3	2.1850	24	666	20.2	19.31
## 451	6.71772	0.0	18.10	0	0.7130	6.749	92.6	2.3236	24	666	20.2	17.44
## 452	5.44114	0.0	18.10	0	0.7130	6.655	98.2	2.3552	24	666	20.2	17.73

## 453	5.09017	0.0	18.10	0	0.7130	6.297	91.8	2.3682	24	666	20.2	17.27
## 454	8.24809	0.0	18.10	0	0.7130	7.393	99.3	2.4527	24	666	20.2	16.74
## 455	9.51363	0.0	18.10	0	0.7130	6.728	94.1	2.4961	24	666	20.2	18.71
## 456	4.75237	0.0	18.10	0	0.7130	6.525	86.5	2.4358	24	666	20.2	18.13
## 457	4.66883	0.0	18.10	0	0.7130	5.976	87.9	2.5806	24	666	20.2	19.01
## 458	8.20058	0.0	18.10	0	0.7130	5.936	80.3	2.7792	24	666	20.2	16.94
## 459	7.75223	0.0	18.10	0	0.7130	6.301	83.7	2.7831	24	666	20.2	16.23
## 460	6.80117	0.0	18.10	0	0.7130	6.081	84.4	2.7175	24	666	20.2	14.70
## 461	4.81213	0.0	18.10	0	0.7130	6.701	90.0	2.5975	24	666	20.2	16.42
## 462	3.69311	0.0	18.10	0	0.7130	6.376	88.4	2.5671	24	666	20.2	14.65
## 463	6.65492	0.0	18.10	0	0.7130	6.317	83.0	2.7344	24	666	20.2	13.99
## 464	5.82115	0.0	18.10	0	0.7130	6.513	89.9	2.8016	24	666	20.2	10.29
## 465	7.83932	0.0	18.10	0	0.6550	6.209	65.4	2.9634	24	666	20.2	13.22
## 466	3.16360	0.0	18.10	0	0.6550	5.759	48.2	3.0665	24	666	20.2	14.13
## 467	3.77498	0.0	18.10	0	0.6550	5.952	84.7	2.8715	24	666	20.2	17.15
## 468	4.42228	0.0	18.10	0	0.5840	6.003	94.5	2.5403	24	666	20.2	21.32
## 469	15.57570	0.0	18.10	0	0.5800	5.926	71.0	2.9084	24	666	20.2	18.13
## 470	13.07510	0.0	18.10	0	0.5800	5.713	56.7	2.8237	24	666	20.2	14.76
## 471	4.34879	0.0	18.10	0	0.5800	6.167	84.0	3.0334	24	666	20.2	16.29
## 472	4.03841	0.0	18.10	0	0.5320	6.229	90.7	3.0993	24	666	20.2	12.87
## 473	3.56868	0.0	18.10	0	0.5800	6.437	75.0	2.8965	24	666	20.2	14.36
## 474	4.64689	0.0	18.10	0	0.6140	6.980	67.6	2.5329	24	666	20.2	11.66
## 475	8.05579	0.0	18.10	0	0.5840	5.427	95.4	2.4298	24	666	20.2	18.14
## 476	6.39312	0.0	18.10	0	0.5840	6.162	97.4	2.2060	24	666	20.2	24.10
## 477	4.87141	0.0	18.10	0	0.6140	6.484	93.6	2.3053	24	666	20.2	18.68
## 478	15.02340	0.0	18.10	0	0.6140	5.304	97.3	2.1007	24	666	20.2	24.91
## 479	10.23300	0.0	18.10	0	0.6140	6.185	96.7	2.1705	24	666	20.2	18.03
## 480	14.33370	0.0	18.10	0	0.6140	6.229	88.0	1.9512	24	666	20.2	13.11
## 481	5.82401	0.0	18.10	0	0.5320	6.242	64.7	3.4242	24	666	20.2	10.74
## 482	5.70818	0.0	18.10	0	0.5320	6.750	74.9	3.3317	24	666	20.2	7.74
## 483	5.73116	0.0	18.10	0	0.5320	7.061	77.0	3.4106	24	666	20.2	7.01
## 484	2.81838	0.0	18.10	0	0.5320	5.762	40.3	4.0983	24	666	20.2	10.42
## 485	2.37857	0.0	18.10	0	0.5830	5.871	41.9	3.7240	24	666	20.2	13.34
## 486	3.67367	0.0	18.10	0	0.5830	6.312	51.9	3.9917	24	666	20.2	10.58
## 487	5.69175	0.0	18.10	0	0.5830	6.114	79.8	3.5459	24	666	20.2	14.98
## 488	4.83567	0.0	18.10	0	0.5830	5.905	53.2	3.1523	24	666	20.2	11.45
## 489	0.15086	0.0	27.74	0	0.6090	5.454	92.7	1.8209	4	711	20.1	18.06
## 490	0.18337	0.0	27.74	0	0.6090	5.414	98.3	1.7554	4	711	20.1	23.97
## 491	0.20746	0.0	27.74	0	0.6090	5.093	98.0	1.8226	4	711	20.1	29.68
## 492	0.10574	0.0	27.74	0	0.6090	5.983	98.8	1.8681	4	711	20.1	18.07
## 493	0.11132	0.0	27.74	0	0.6090	5.983	83.5	2.1099	4	711	20.1	13.35
## 494	0.17331	0.0	9.69	0	0.5850	5.707	54.0	2.3817	6	391	19.2	12.01
## 495	0.27957	0.0	9.69	0	0.5850	5.926	42.6	2.3817	6	391	19.2	13.59
## 496	0.17899	0.0	9.69	0	0.5850	5.670	28.8	2.7986	6	391	19.2	17.60
## 497	0.28960	0.0	9.69	0	0.5850	5.390	72.9	2.7986	6	391	19.2	21.14
## 498	0.26838	0.0	9.69	0	0.5850	5.794	70.6	2.8927	6	391	19.2	14.10
## 499	0.23912	0.0	9.69	0	0.5850	6.019	65.3	2.4091	6	391	19.2	12.92
## 500	0.17783	0.0	9.69	0	0.5850	5.569	73.5	2.3999	6	391	19.2	15.10
## 501	0.22438	0.0	9.69	0	0.5850	6.027	79.7	2.4982	6	391	19.2	14.33
## 502	0.06263	0.0	11.93	0	0.5730	6.593	69.1	2.4786	1	273	21.0	9.67
## 503	0.04527	0.0	11.93	0	0.5730	6.120	76.7	2.2875	1	273	21.0	9.08
## 504	0.06076	0.0	11.93	0	0.5730	6.976	91.0	2.1675	1	273	21.0	5.64
## 505	0.10959	0.0	11.93	0	0.5730	6.794	89.3	2.3889	1	273	21.0	6.48
## 506	0.04741	0.0	11.93	0	0.5730	6.030	80.8	2.5050	1	273	21.0	7.88

##	medv
## 1	24.0
## 2	21.6
## 3	34.7
## 4	33.4
## 5	36.2
## 6	28.7
## 7	22.9
## 8	27.1
## 9	16.5
## 10	18.9
## 11	15.0
## 12	18.9
## 13	21.7
## 14	20.4
## 15	18.2
## 16	19.9
## 17	23.1
## 18	17.5
## 19	20.2
## 20	18.2
## 21	13.6
## 22	19.6
## 23	15.2
## 24	14.5
## 25	15.6
## 26	13.9
## 27	16.6
## 28	14.8
## 29	18.4
## 30	21.0
## 31	12.7
## 32	14.5
## 33	13.2
## 34	13.1
## 35	13.5
## 36	18.9
## 37	20.0
## 38	21.0
## 39	24.7
## 40	30.8
## 41	34.9
## 42	26.6
## 43	25.3
## 44	24.7
## 45	21.2
## 46	19.3
## 47	20.0
## 48	16.6
## 49	14.4
## 50	19.4
## 51	19.7
## 52	20.5
## 53	25.0

##	54	23.4
##	55	18.9
##	56	35.4
##	57	24.7
##	58	31.6
##	59	23.3
##	60	19.6
##	61	18.7
##	62	16.0
##	63	22.2
##	64	25.0
##	65	33.0
##	66	23.5
##	67	19.4
##	68	22.0
##	69	17.4
##	70	20.9
##	71	24.2
##	72	21.7
##	73	22.8
##	74	23.4
##	75	24.1
##	76	21.4
##	77	20.0
##	78	20.8
##	79	21.2
##	80	20.3
##	81	28.0
##	82	23.9
##	83	24.8
##	84	22.9
##	85	23.9
##	86	26.6
##	87	22.5
##	88	22.2
##	89	23.6
##	90	28.7
##	91	22.6
##	92	22.0
##	93	22.9
##	94	25.0
##	95	20.6
##	96	28.4
##	97	21.4
##	98	38.7
##	99	43.8
##	100	33.2
##	101	27.5
##	102	26.5
##	103	18.6
##	104	19.3
##	105	20.1
##	106	19.5
##	107	19.5

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176 29.4
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182 36.2
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197 33.3
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199 34.6
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201 32.9
202 24.1
203 42.3
204 48.5
205 50.0
206 22.6
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349 24.5
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368 23.1
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370 50.0
371 50.0
372 50.0
373 50.0
374 13.8
375 13.8
376 15.0
377 13.9

378 13.3
379 13.1
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381 10.4
382 10.9
383 11.3
384 12.3
385 8.8
386 7.2
387 10.5
388 7.4
389 10.2
390 11.5
391 15.1
392 23.2
393 9.7
394 13.8
395 12.7
396 13.1
397 12.5
398 8.5
399 5.0
400 6.3
401 5.6
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403 12.1
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406 5.0
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409 17.2
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414 16.3
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417 7.5
418 10.4
419 8.8
420 8.4
421 16.7
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423 20.8
424 13.4
425 11.7
426 8.3
427 10.2
428 10.9
429 11.0
430 9.5
431 14.5

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436 13.4
437 9.6
438 8.7
439 8.4
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441 10.5
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443 18.4
444 15.4
445 10.8
446 11.8
447 14.9
448 12.6
449 14.1
450 13.0
451 13.4
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455 14.9
456 14.1
457 12.7
458 13.5
459 14.9
460 20.0
461 16.4
462 17.7
463 19.5
464 20.2
465 21.4
466 19.9
467 19.0
468 19.1
469 19.1
470 20.1
471 19.9
472 19.6
473 23.2
474 29.8
475 13.8
476 13.3
477 16.7
478 12.0
479 14.6
480 21.4
481 23.0
482 23.7
483 25.0
484 21.8
485 20.6

```
## 486 21.2
## 487 19.1
## 488 20.6
## 489 15.2
## 490 7.0
## 491 8.1
## 492 13.6
## 493 20.1
## 494 21.8
## 495 24.5
## 496 23.1
## 497 19.7
## 498 18.3
## 499 21.2
## 500 17.5
## 501 16.8
## 502 22.4
## 503 20.6
## 504 23.9
## 505 22.0
## 506 11.9
```

Reading about the data set:

```
?Boston
```

How many rows are in this data set? How many columns? What do the rows and columns represent?

Answer: This data set has 506 rows and 13 columns. Rows represent 506 suburbs of Boston and Columns represent different variables from suburbs like crim = crime, zn = proportion of residential land, indus = proportion of non-retail business, etc.

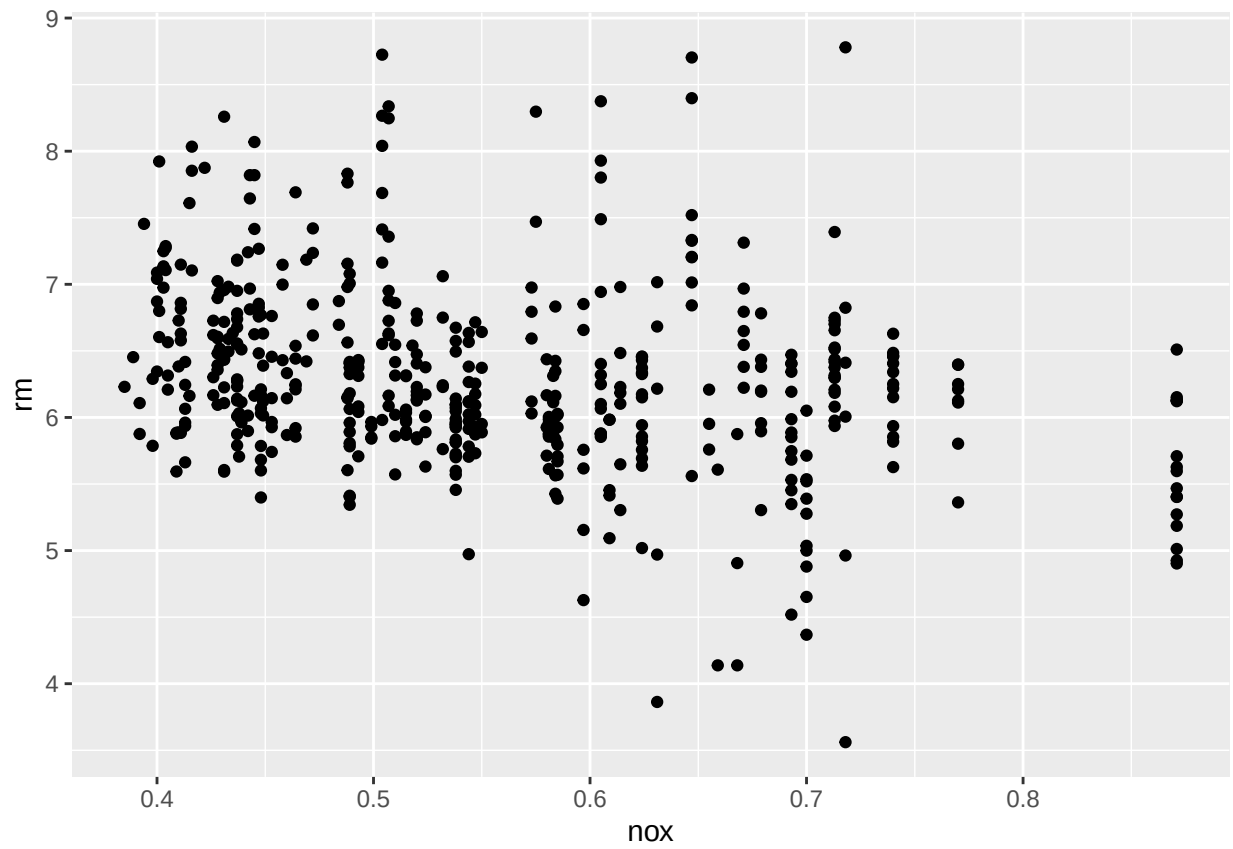
```
dim(Boston)
```

```
## [1] 506 13
```

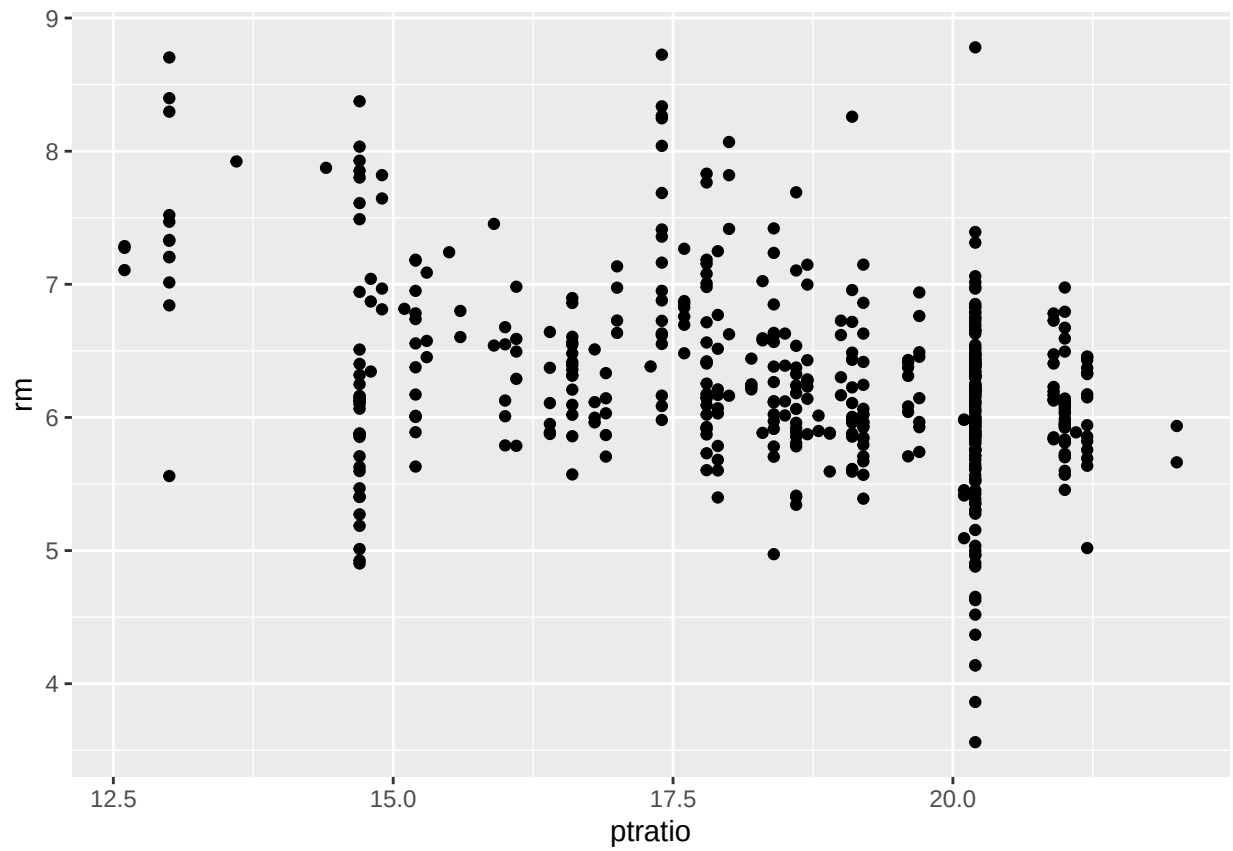
(b) Make some pairwise scatterplots of the predictors (columns) in this data set. Describe your findings.

Answer:

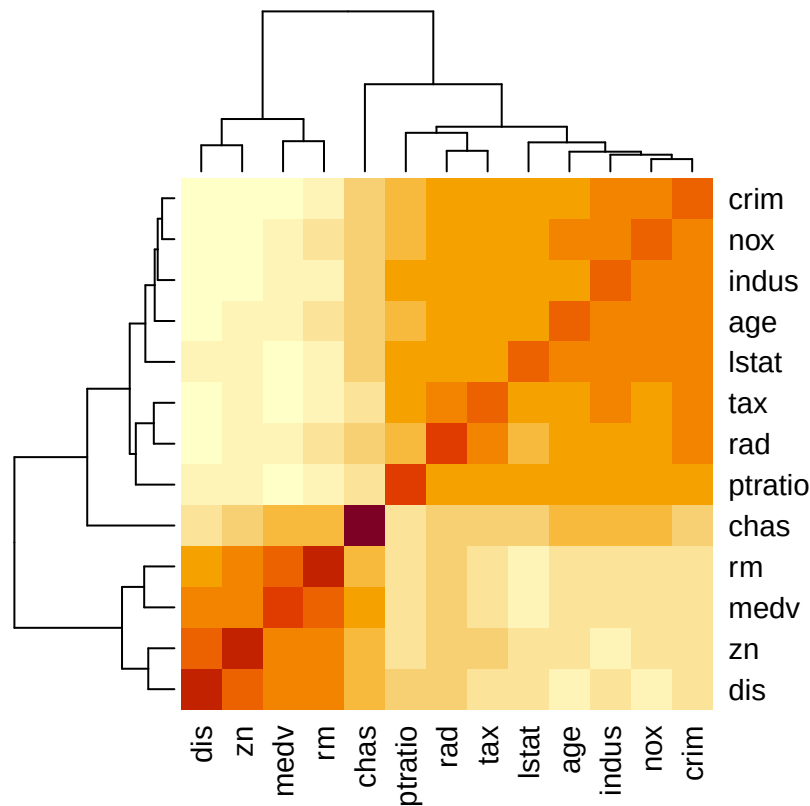
```
ggplot(Boston, aes(nox, rm)) +
  geom_point()
```



```
ggplot(Boston, aes(nox, rm)) +  
  geom_point()
```



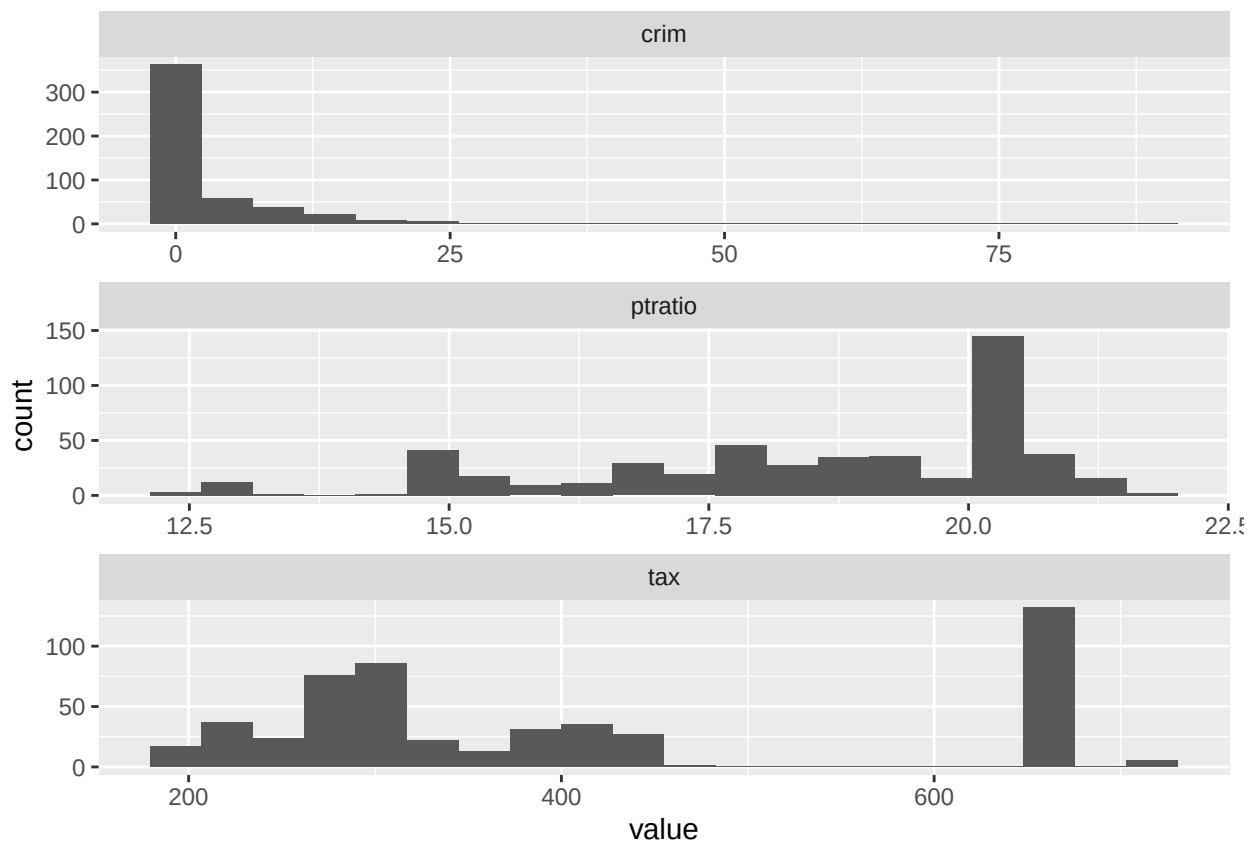
```
heatmap(cor(Boston, method = "spearman"), cexRow = 1.1, cexCol = 1.1)
```



- (c) Are any of the predictors associated with per capita crime rate? If so, explain the relationship. Answer: Yes Relationship: there is a relation between crime rate and radial highways. We can see crime rate increase with radial highways. We can also notice the increase in crime rate with proportion of owner-occupied units built prior to 1940.
- (d) Do any of the census tracts of Boston appear to have particularly high crime rates? Tax rates? Pupil-teacher ratios? Comment on the range of each predictor.

Answer: Yes—there are census tracts with particularly high crime and high tax rates (clear outliers). Pupil-teacher ratios vary too, but much less dramatically.

```
Boston |>
  pivot_longer(cols = 1:13) |>
  filter(name %in% c("crim", "tax", "ptratio")) |>
  ggplot(aes(value)) +
  geom_histogram(bins = 20) +
  facet_wrap(~name, scales = "free", ncol = 1)
```



(e) How many of the census tracts in this data set bound the Charles river?

Answer: There are 35 suburbs that bound the Charles river.

```
sum(Boston$chas)
```

```
## [1] 35
```

(f) What is the median pupil-teacher ratio among the towns in this data set?

Answer: The median is 19.05

```
median(Boston$ptratio)
```

```
## [1] 19.05
```

(g) Which census tract of Boston has lowest median value of owner-occupied homes? What are the values of the other predictors for that census tract, and how do those values compare to the overall ranges for those predictors? Comment on your findings.

Answer: The suburb 399 and 406 has the lowest median value of owner-occupied homes in \$1000s

```
Boston[Boston$medv == min(Boston$medv), ] |>
  kable()
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	lstat	medv
399	38.3518	0	18.1	0	0.693	5.453	100	1.4896	24	666	20.2	30.59	5
406	67.9208	0	18.1	0	0.693	5.683	100	1.4254	24	666	20.2	22.98	5

```
sapply(Boston, quantile) |>
  kable()
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	lstat	medv
0%	0.006320	0.0	0.46	0	0.385	3.5610	2.900	1.129600	1	187	12.60	1.730	5.000
25%	0.082045	0.0	5.19	0	0.449	5.8855	45.025	2.100175	4	279	17.40	6.950	17.025
50%	0.256510	0.0	9.69	0	0.538	6.2085	77.500	3.207450	5	330	19.05	11.360	21.200
75%	3.677083	12.5	18.10	0	0.624	6.6235	94.075	5.188425	24	666	20.20	16.955	25.000
100%	88.976200	100.0	27.74	1	0.871	8.7800	100.000	12.126500	24	711	22.00	37.970	50.000

Comments on findings: The minimum medv is 5.0 (in \$1,000s).

1. That tract's other predictors sit at extreme, unfavorable ends of many ranges:
 2. lstat very high (near the top of its range) → many lower-status households.
 - a. rm low → small homes.
 - b. nox high, dis low, indus high → dense/industrial, close-in, more pollution.
 - c. rad very high and tax high → near major highways with high property-tax rate.
 - d. ptratio high → more students per teacher.
 - e. crim typically high in this tract as well.
 3. Overall, the tract with the lowest medv lies at (or near) the worst ends of several predictors' ranges, which aligns with lower house values (fewer rooms, more pollution/traffic/industry, higher tax and pupil-teacher ratios, higher % lower-status).
- (h) In this data set, how many of the census tracts average more than seven rooms per dwelling? More than eight rooms per dwelling? Comment on the census tracts that average more than eight rooms per dwelling.

Answer: There are 64 census tracts with more than 7 rm. There are 13 tracts with more than 8 rm. (rm = rooms per dwelling)

```
sum(Boston$rm > 7)
```

```
## [1] 64
```

```
sum(Boston$rm > 8)
```

```
## [1] 13
```

Let's compare median statistics for those census tracts with more than eight rooms per dwelling on average, with the statistics for those with fewer.

```

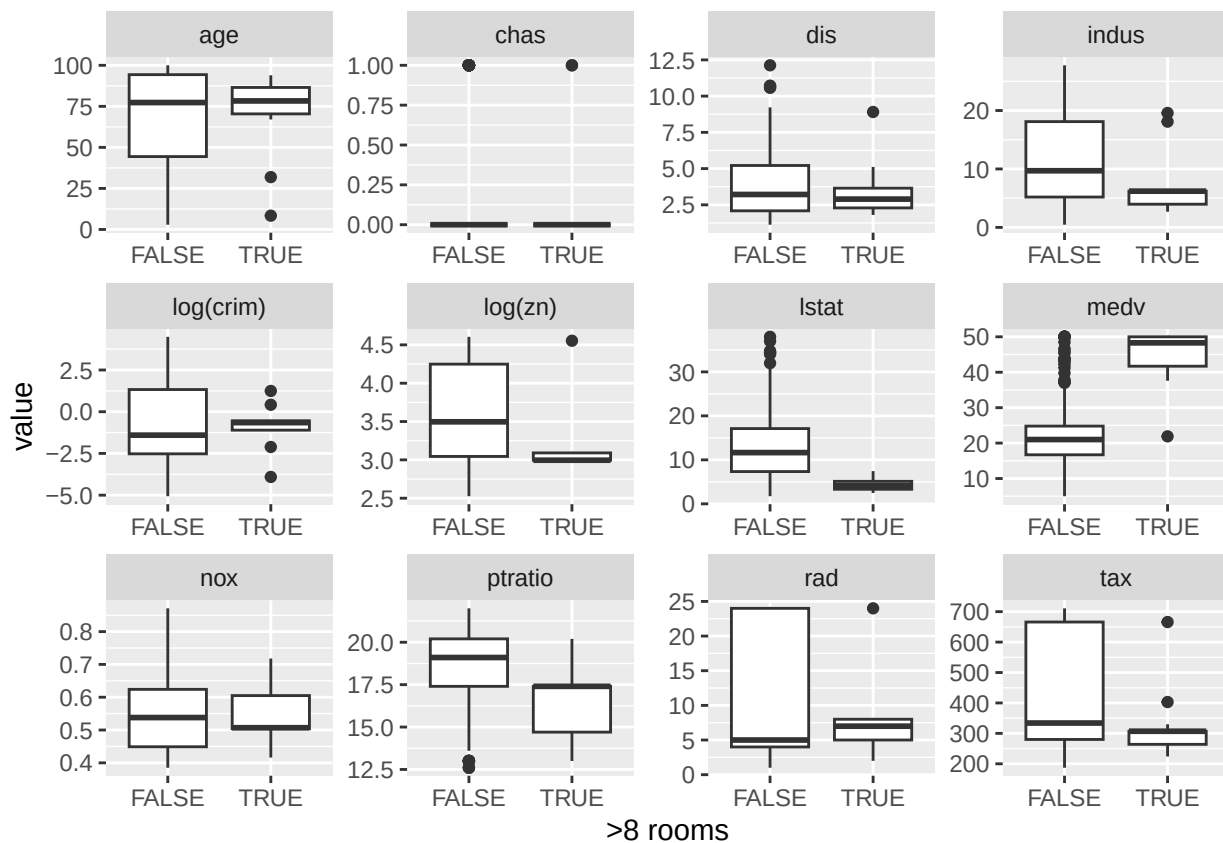
Boston |>
  mutate(
    `log(crim)` = log(crim),
    `log(zn)` = log(zn)
  ) |>
  select(-c(crim, zn)) |>
  pivot_longer(!rm) |>
  mutate(">8 rooms" = rm > 8) |>
  ggplot(aes(`>8 rooms`, value)) +
  geom_boxplot() +
  facet_wrap(~name, scales = "free")

```

```

## Warning: Removed 372 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

```



Census tracts with big average properties (more than eight rooms per dwelling) have higher median value (**medv**), a lower proportion of non-retail business acres (**indus**), a lower pupil-teacher ratio (**ptratio**), a lower status of the population (**lstat**) among other differences.

Section 3.7: Question 04

- I collect a set of data ($n = 100$ observations) containing a single predictor and a quantitative response. I then fit a linear regression model to the data, as well as a separate cubic regression, i.e. $Y = 0 + 1X + 2X^2 + 3X^3 + \dots$

- (a) Suppose that the true relationship between X and Y is linear, i.e. $Y = \beta_0 + \beta_1 X + \epsilon$. Consider the training residual sum of squares (RSS) for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.

Answer: You would expect the cubic regression to have lower RSS than linear regression. Reason: Ordinary least squares minimizes residual sum of squares, and the cubic model nests the linear model (adds x^2, x^3), so training residual sum of squares can't increase.

- (b) Answer (a) using test rather than training RSS. Answer: Though we could not be certain, the test RSS would likely be higher due to overfitting.
- (c) Suppose that the true relationship between X and Y is not linear, but we don't know how far it is from linear. Consider the training RSS for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.

Answer: Cubic residual sum of squares $\leq$ Linear residual sum of squares. Reason: Same nesting argument—more flexible model fits training data at least as well.

- (d) Answer (c) using test rather than training RSS. Answer: Not enough information. Reason: Outcome depends on bias–variance trade-off: cubic may lower bias (captures curvature) but raises variance; depends on degree of nonlinearity, noise level, and sample size.

Section 3.7: Question 15

Q. This problem involves the `Boston` data set, which we saw in the lab for this chapter. We will now try to predict per capita crime rate using the other variables in this data set. In other words, per capita crime rate is the response, and the other variables are the predictors.

We are trying to predict `crim`.

```
pred <- subset(Boston, select = -crim)
```

- a. For each predictor, fit a simple linear regression model to predict the response. Describe your results. In which of the models is there a statistically significant association between the predictor and the response? Create some plots to back up your assertions.

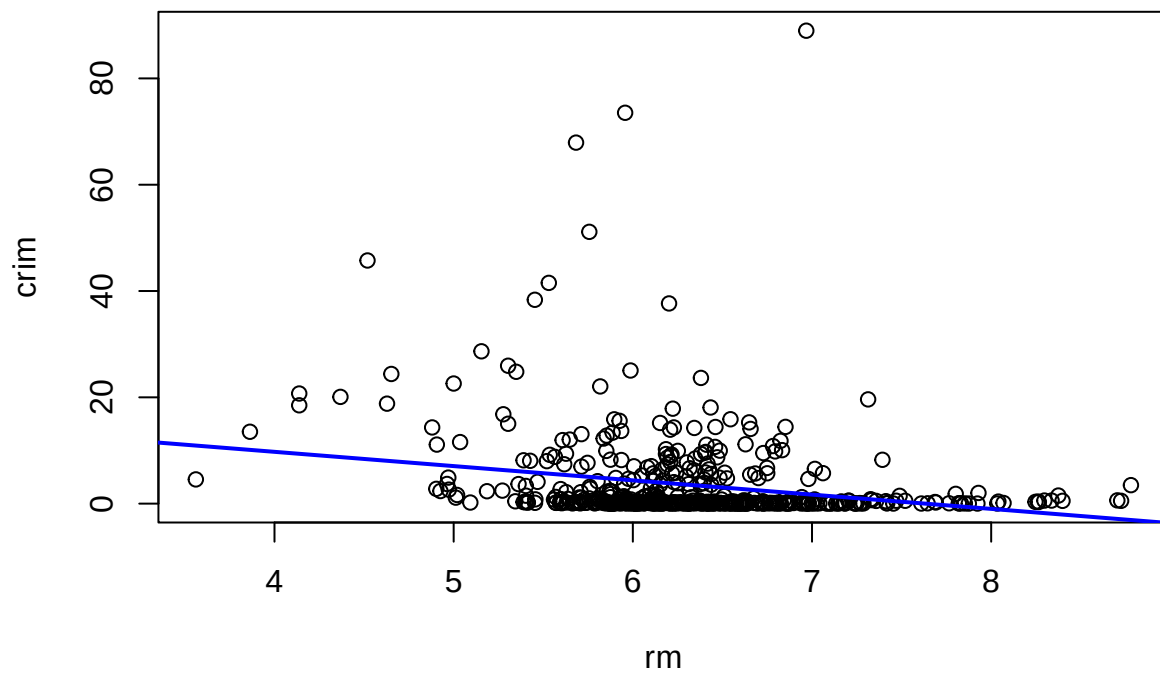
```
fits <- lapply(pred, function(x) lm(Boston$crim ~ x))
printCoefmat(do.call(rbind, lapply(fits, function(x) coef(summary(x))[2, ])))
```

##		Estimate	Std. Error	t value	Pr(> t)
##	zn	-0.0739350	0.0160946	-4.5938	5.506e-06 ***
##	indus	0.5097763	0.0510243	9.9908	< 2.2e-16 ***
##	chas	-1.8927766	1.5061155	-1.2567	0.2094
##	nox	31.2485312	2.9991904	10.4190	< 2.2e-16 ***
##	rm	-2.6840512	0.5320411	-5.0448	6.347e-07 ***
##	age	0.1077862	0.0127364	8.4628	2.855e-16 ***
##	dis	-1.5509017	0.1683300	-9.2135	< 2.2e-16 ***
##	rad	0.6179109	0.0343318	17.9982	< 2.2e-16 ***
##	tax	0.0297423	0.0018474	16.0994	< 2.2e-16 ***
##	ptratio	1.1519828	0.1693736	6.8014	2.943e-11 ***

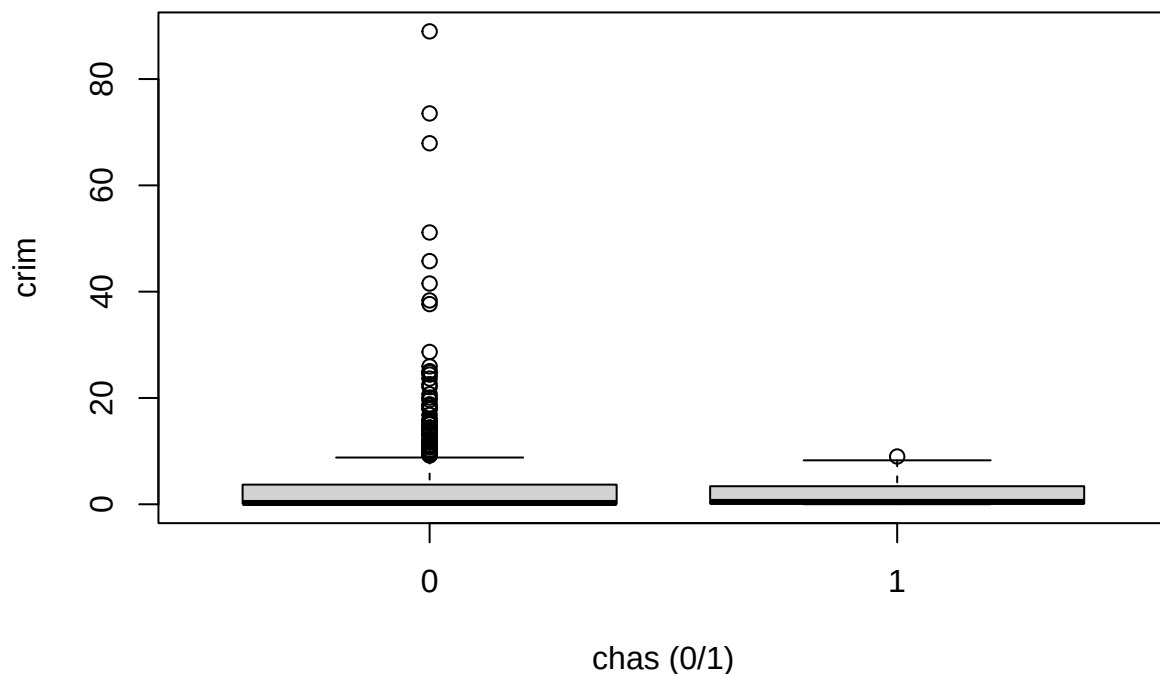
```
## lstat    0.5488048  0.0477610 11.4907 < 2.2e-16 ***
## medv    -0.3631599  0.0383902 -9.4597 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

There are significant associations for all predictors with the exception of `chas` when fitting separate linear models. For example, consider the following plot representing the third model

```
# Example 1: continuous predictor (rm)
plot(Boston$rm, Boston$crim, xlab = "rm", ylab = "crim")
abline(fits[["rm"]], col = "blue", lwd = 2)
```



```
# Example 2: binary predictor (chas) - the non-significant one
boxplot(crim ~ chas, data = Boston, xlab = "chas (0/1)", ylab = "crim")
```



- b. Fit a multiple regression model to predict the response using all of the predictors. Describe your results. For which predictors can we reject the null hypothesis $H_0 : \beta_j = 0$?

```
mfit <- lm(crim ~ ., data = Boston)
summary(mfit)
```

```
##
## Call:
## lm(formula = crim ~ ., data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.534 -2.248 -0.348  1.087 73.923
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 13.7783938   7.0818258   1.946 0.052271 .
## zn           0.0457100   0.0187903   2.433 0.015344 *
## indus       -0.0583501   0.0836351  -0.698 0.485709
## chas        -0.8253776   1.1833963  -0.697 0.485841
## nox        -9.9575865   5.2898242  -1.882 0.060370 .
## rm           0.6289107   0.6070924   1.036 0.300738
## age        -0.0008483   0.0179482  -0.047 0.962323
## dis        -1.0122467   0.2824676  -3.584 0.000373 ***
## rad          0.6124653   0.0875358   6.997 8.59e-12 ***
```

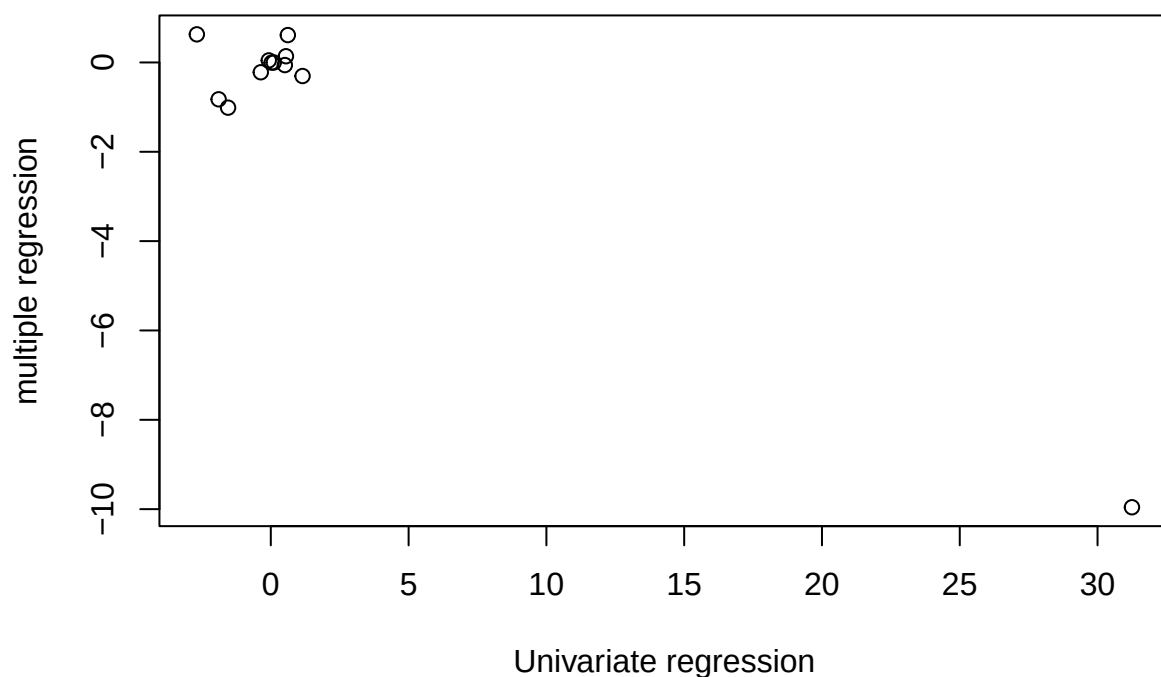
```
## tax          -0.0037756  0.0051723  -0.730  0.465757
## ptratio      -0.3040728  0.1863598  -1.632  0.103393
## lstat         0.1388006  0.0757213   1.833  0.067398 .
## medv         -0.2200564  0.0598240  -3.678  0.000261 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.46 on 493 degrees of freedom
## Multiple R-squared:  0.4493, Adjusted R-squared:  0.4359
## F-statistic: 33.52 on 12 and 493 DF,  p-value: < 2.2e-16
```

There are now only significant associations for zn, dis, rad, black and medv.

- c. How do your results from (a) compare to your results from (b)? Create a plot displaying the univariate regression coefficients from (a) on the x -axis, and the multiple regression coefficients from (b) on the y -axis. That is, each predictor is displayed as a single point in the plot. Its coefficient in a simple linear regression model is shown on the x -axis, and its coefficient estimate in the multiple linear regression model is shown on the y -axis.

The results from (b) show reduced significance compared to the models fit in (a).

```
plot(sapply(fits, function(x) coef(x)[2]), coef(mfit)[-1],
     xlab = "Univariate regression",
     ylab = "multiple regression"
)
```



The estimated coefficients differ (in particular the estimated coefficient for `nox` is dramatically different) between the two modelling strategies.

- d. Is there evidence of non-linear association between any of the predictors and the response? To answer this question, for each predictor X , fit a model of the form $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$

Answer:

```
pred <- subset(pred, select = -chas)
fits <- lapply(names(pred), function(p) {
  f <- paste0("crim ~ poly(", p, ", 3)")
  lm(as.formula(f), data = Boston)
})
for (fit in fits) printCoefmat(coef(summary(fit)))
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.61352    0.37219   9.7088 < 2.2e-16 ***
## poly(zn, 3)1 -38.74984    8.37221  -4.6284 4.698e-06 ***
## poly(zn, 3)2  23.93983    8.37221   2.8594 0.004421 **
## poly(zn, 3)3 -10.07187    8.37221  -1.2030 0.229539
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)     3.6135     0.3300 10.9501 < 2.2e-16 ***
## poly(indus, 3)1  78.5908     7.4231 10.5873 < 2.2e-16 ***
## poly(indus, 3)2 -24.3948     7.4231  -3.2863 0.001086 **
## poly(indus, 3)3 -54.1298     7.4231 -7.2920 1.196e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)     3.61352    0.32157 11.2370 < 2.2e-16 ***
## poly(nox, 3)1   81.37202     7.23361 11.2492 < 2.2e-16 ***
## poly(nox, 3)2 -28.82859     7.23361  -3.9854 7.737e-05 ***
## poly(nox, 3)3 -60.36189     7.23361  -8.3446 6.961e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)     3.6135     0.3703   9.7584 < 2.2e-16 ***
## poly(rm, 3)1  -42.3794     8.3297  -5.0878 5.128e-07 ***
## poly(rm, 3)2   26.5768     8.3297   3.1906 0.001509 **
## poly(rm, 3)3   -5.5103     8.3297  -0.6615 0.508575
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)     3.61352    0.34852 10.3683 < 2.2e-16 ***
## poly(age, 3)1   68.18201     7.83970   8.6970 < 2.2e-16 ***
## poly(age, 3)2   37.48447     7.83970   4.7814 2.291e-06 ***
## poly(age, 3)3   21.35321     7.83970   2.7237 0.00668 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)     3.61352    0.32592 11.0870 < 2.2e-16 ***
## poly(dis, 3)1  -73.38859     7.33148 -10.0101 < 2.2e-16 ***
```

```

## poly(dis, 3)2 56.37304    7.33148    7.6892 7.870e-14 ***
## poly(dis, 3)3 -42.62188    7.33148   -5.8135 1.089e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.61352    0.29707 12.1639 < 2.2e-16 ***
## poly(rad, 3)1 120.90745    6.68240 18.0934 < 2.2e-16 ***
## poly(rad, 3)2  17.49230    6.68240  2.6177 0.009121 **
## poly(rad, 3)3   4.69846    6.68240  0.7031 0.482314
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.61352    0.30468 11.8599 < 2.2e-16 ***
## poly(tax, 3)1 112.64583    6.85371 16.4358 < 2.2e-16 ***
## poly(tax, 3)2  32.08725    6.85371  4.6817 3.665e-06 ***
## poly(tax, 3)3  -7.99681    6.85371 -1.1668 0.2439
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.61352    0.36105 10.0084 < 2.2e-16 ***
## poly(ptratio, 3)1 56.04523    8.12158  6.9008 1.565e-11 ***
## poly(ptratio, 3)2 24.77482    8.12158  3.0505 0.002405 **
## poly(ptratio, 3)3 -22.27974    8.12158 -2.7433 0.006301 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.61352    0.33917 10.6540 <2e-16 ***
## poly(lstat, 3)1 88.06967    7.62944 11.5434 <2e-16 ***
## poly(lstat, 3)2 15.88816    7.62944  2.0825 0.0378 *
## poly(lstat, 3)3 -11.57402    7.62944 -1.5170 0.1299
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.61352    0.29203 12.374 < 2.2e-16 ***
## poly(medv, 3)1 -75.05761    6.56915 -11.426 < 2.2e-16 ***
## poly(medv, 3)2 88.08621    6.56915 13.409 < 2.2e-16 ***
## poly(medv, 3)3 -48.03343    6.56915 -7.312 1.047e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Yes there is strong evidence for many variables having non-linear associations. In many cases, the addition of a cubic term is significant (*indus*, *nox*, *age*, *dis*, *ptratio* and *medv*). In other cases although the cubic terms is not significant, the squared term is (*zn*, *rm*, *rad*, *tax*, *lstat*). In only one case, *black* is there no evidence for a non-linear relationship.